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Luo et al.

(54) **FASTENING CLIP AND FASTENING CLIP ASSEMBLY COMPRISING SAME**

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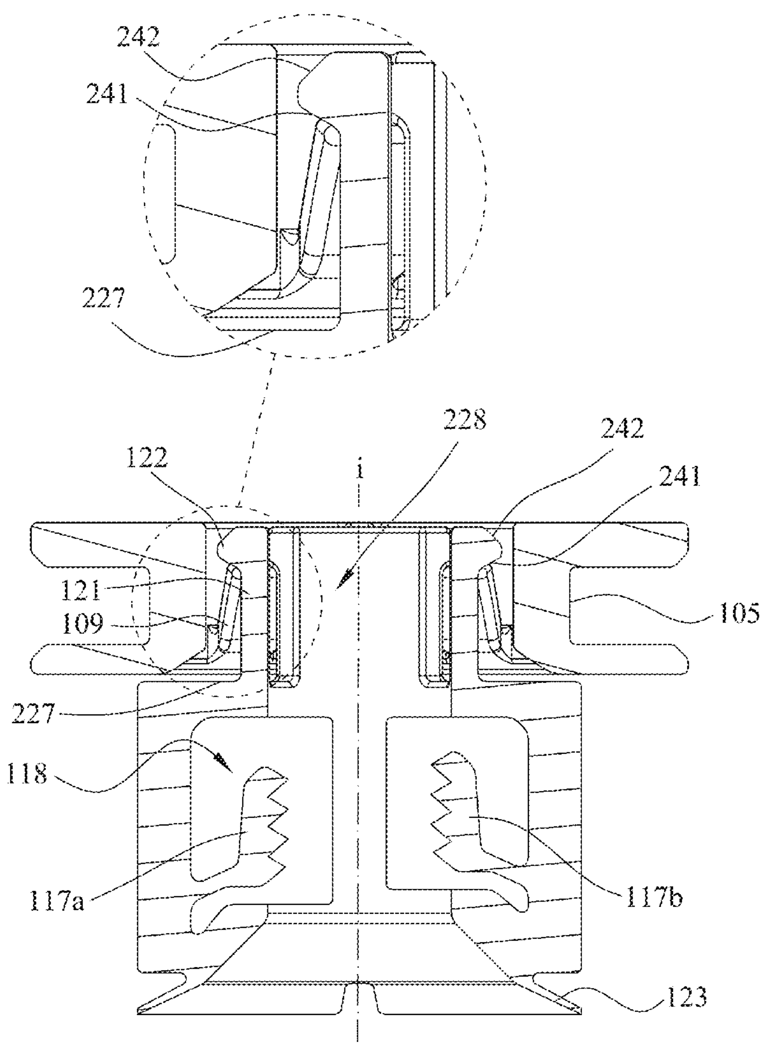
CPC ***F16B 21/073*** (2013.01)

(57) **ABSTRACT**

The present disclosure provides a fastening clip including a first connecter and a second connecter releasably connected to each other. When the fastening clip connects two components, the first connecter and the second connecter are firmly connected to the two components, respectively. Moreover, the two components are separated from each other by releasing the first connecter and the second connecter of the fastening clip of the present disclosure, so that it is possible to avoid wear between the components made of a metal material and the fastening clip made of a plastic material. The fastening clip is particularly suitable for mounting two components with at least one being made of a metal material.



A-A

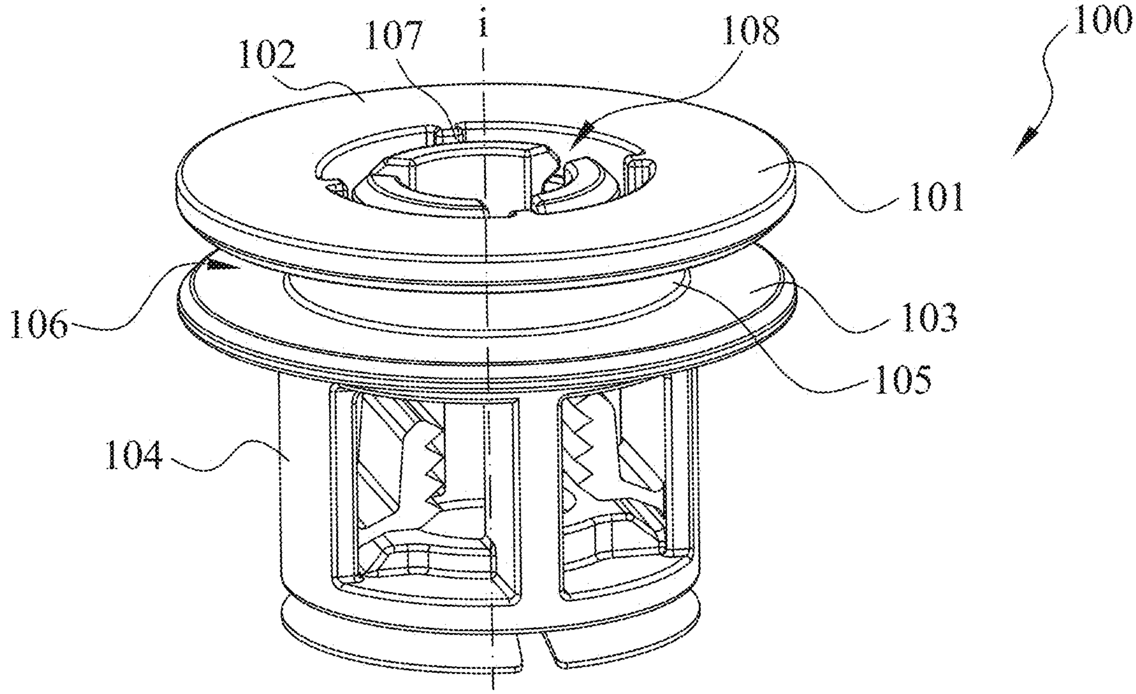

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Fig. 1A

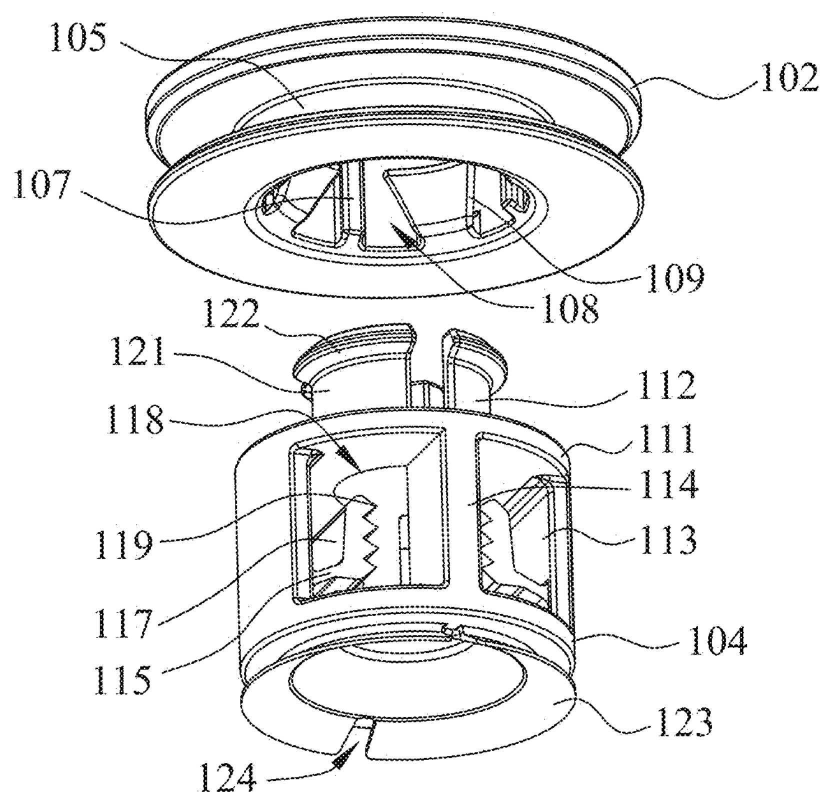
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Fig. 1B

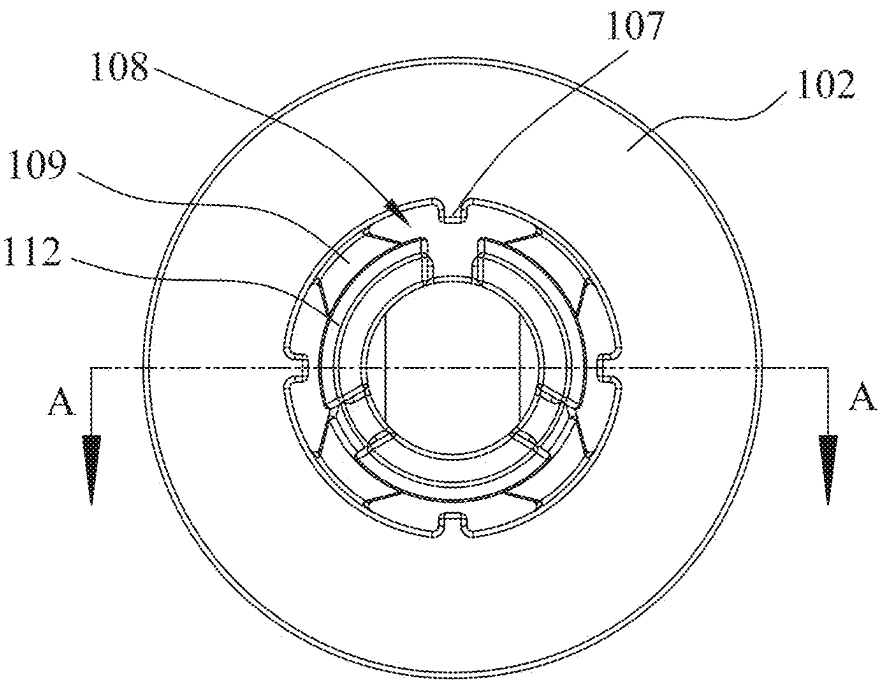


Fig. 1C

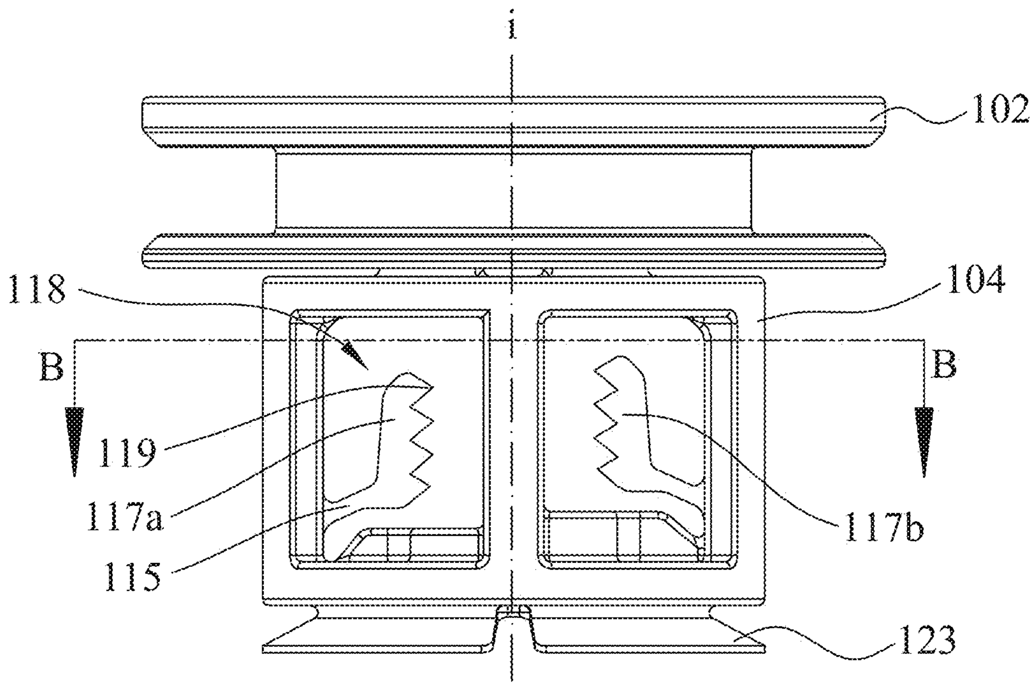
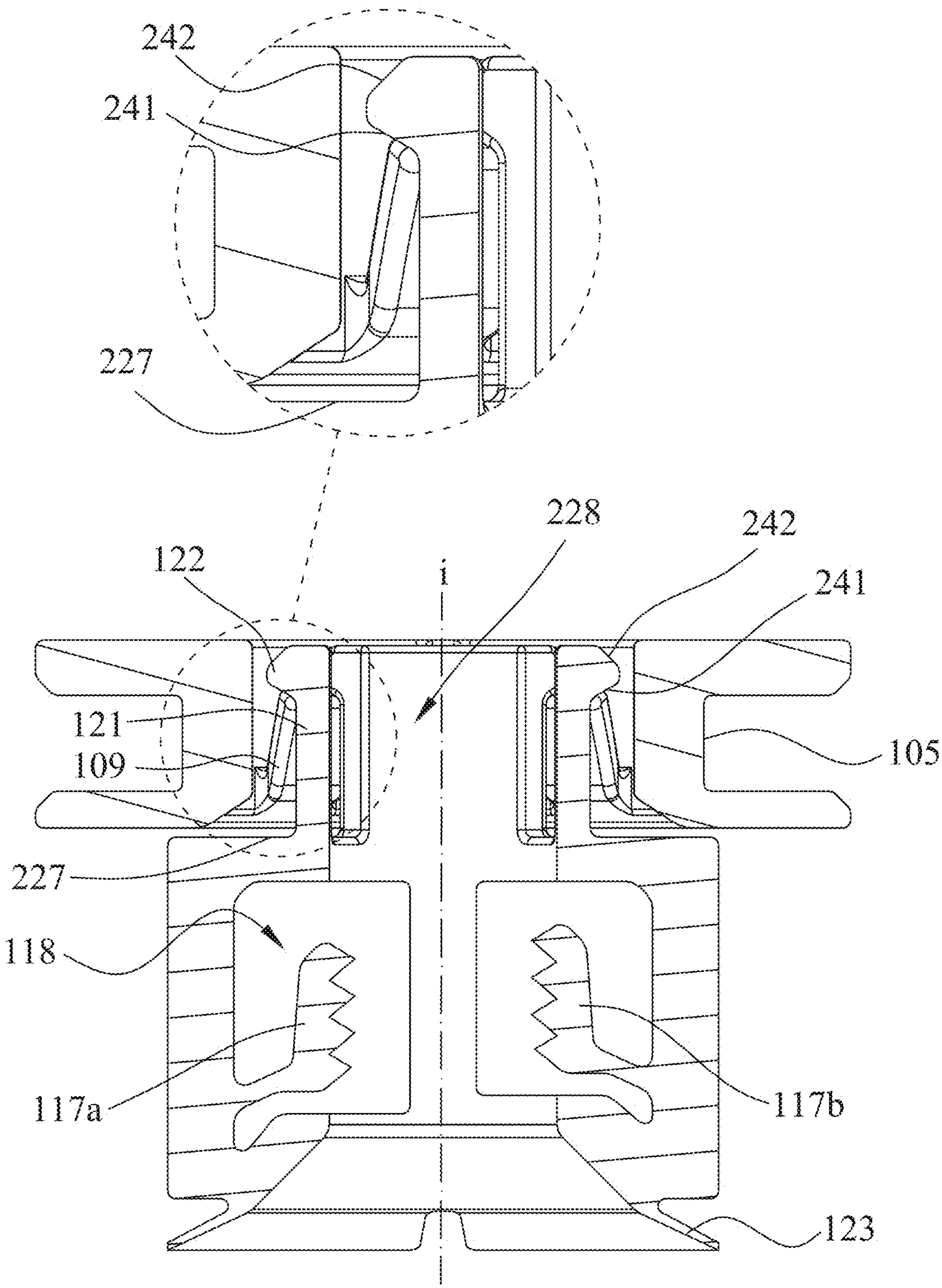


Fig. 1D



A-A

Fig. 2A

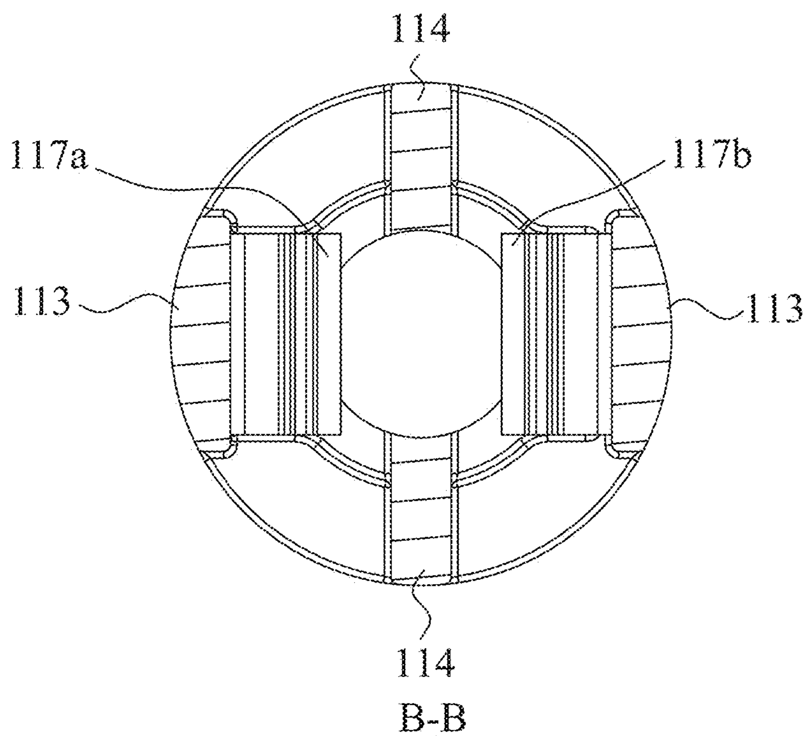


Fig. 2B

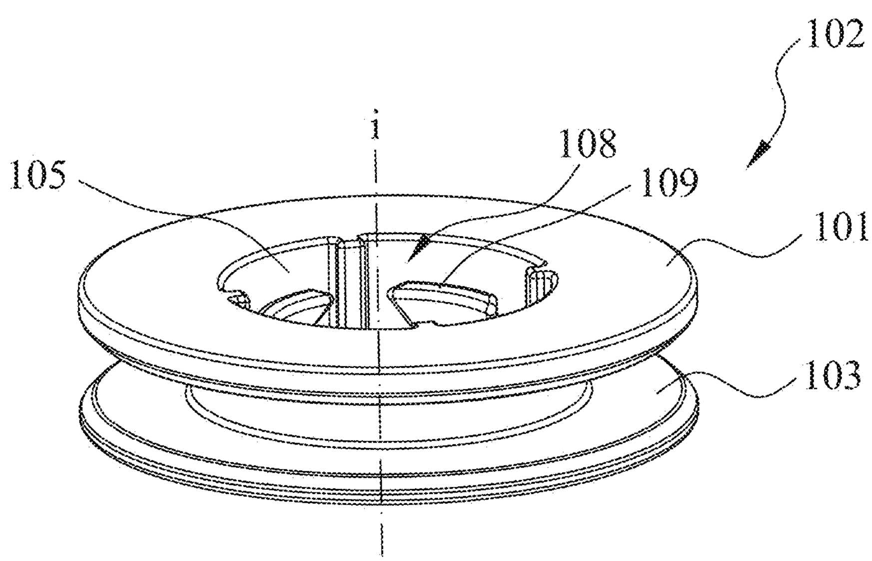


Fig. 3A

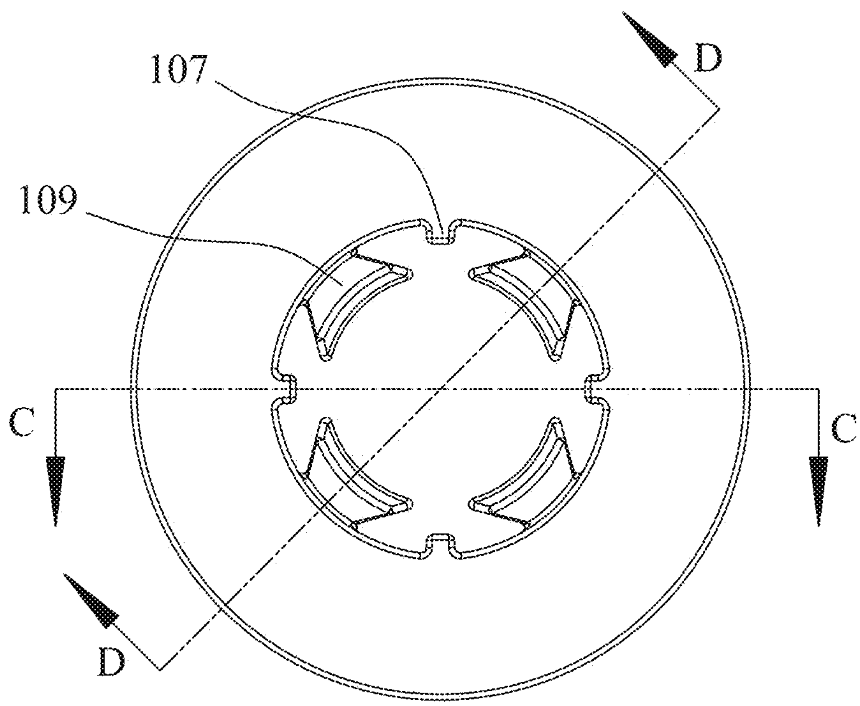
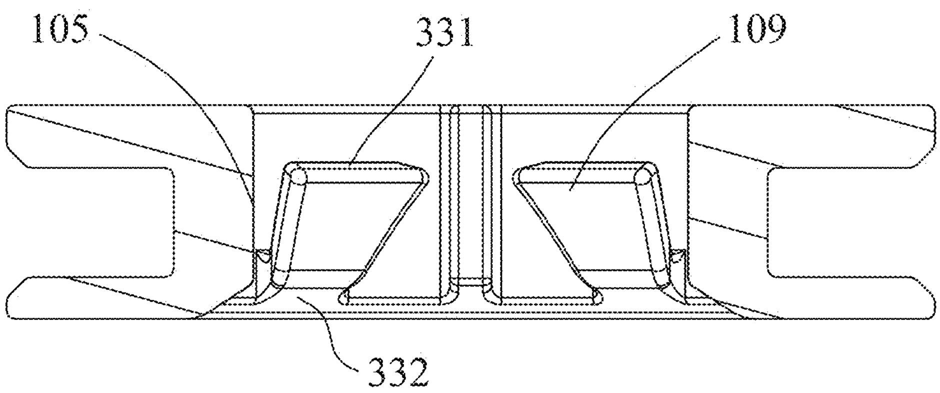
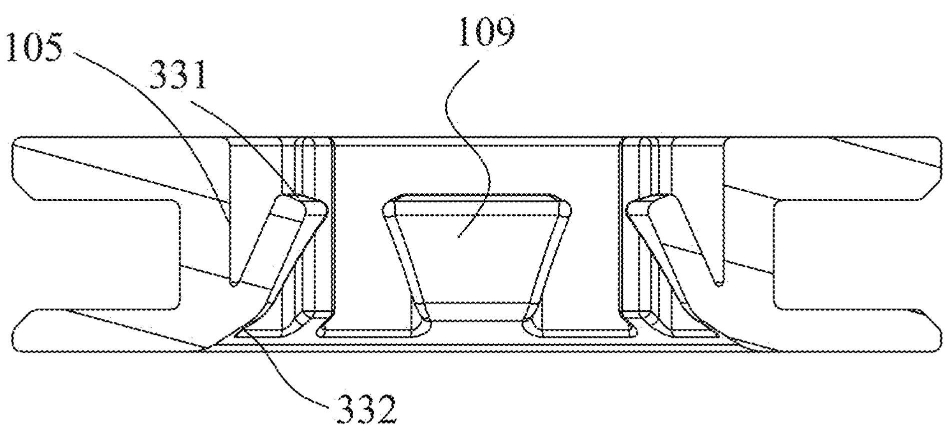


Fig. 3B



C-C

Fig. 3C



D-D

Fig. 3D

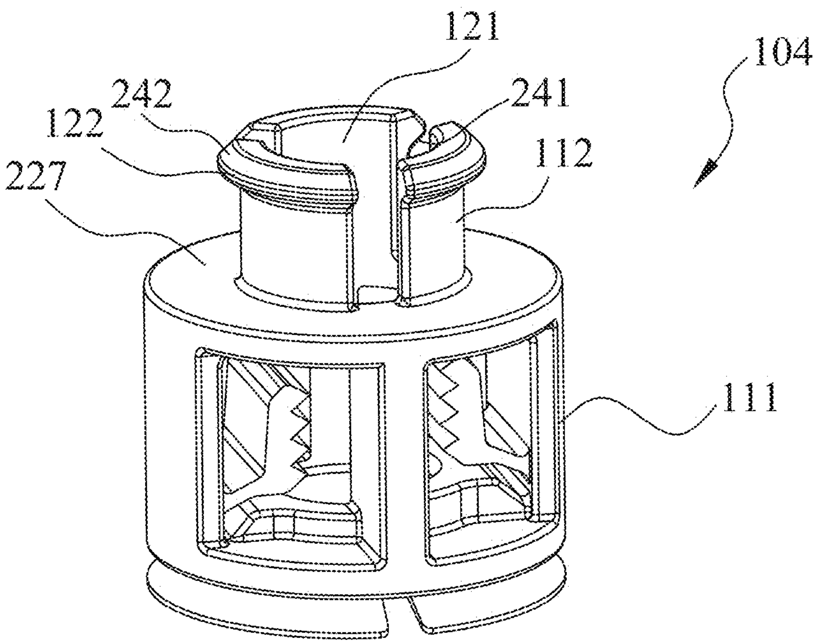


Fig. 4A

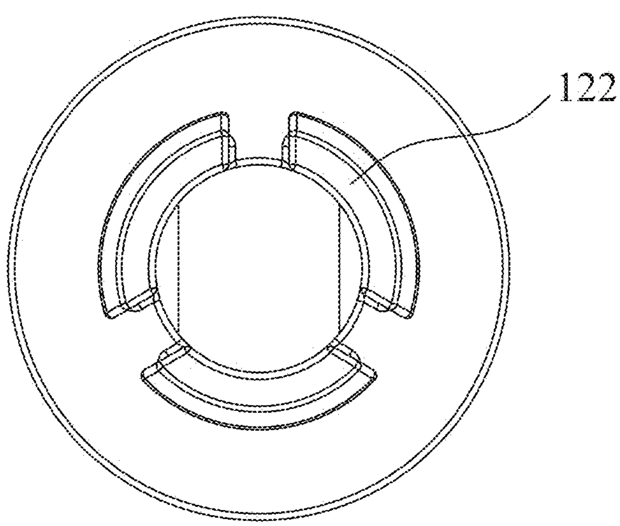


Fig. 4B

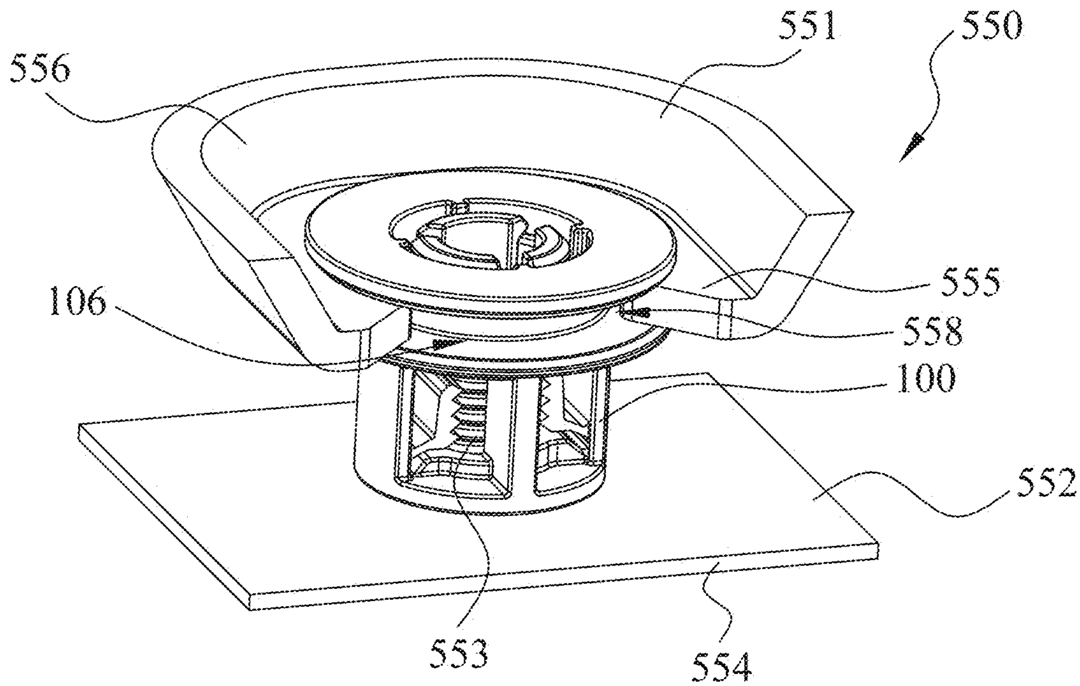

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Fig. 5A

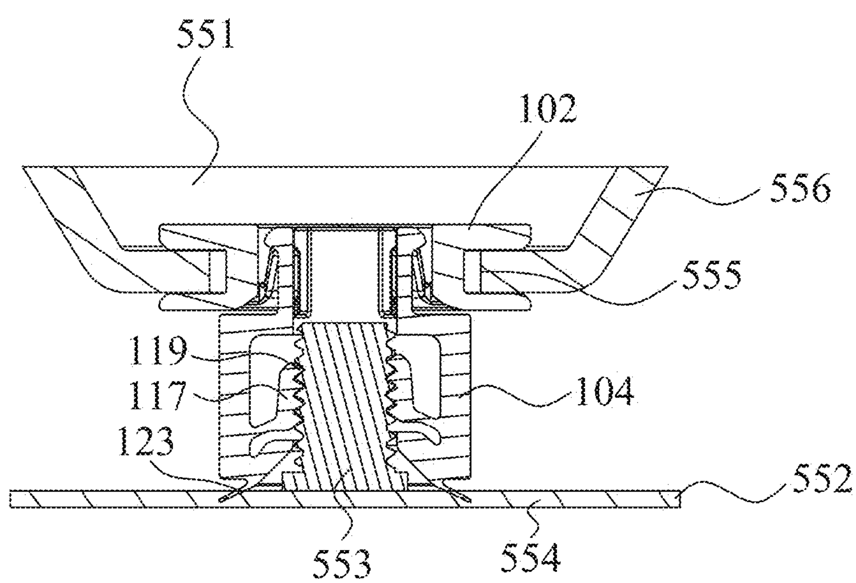
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Fig. 5B

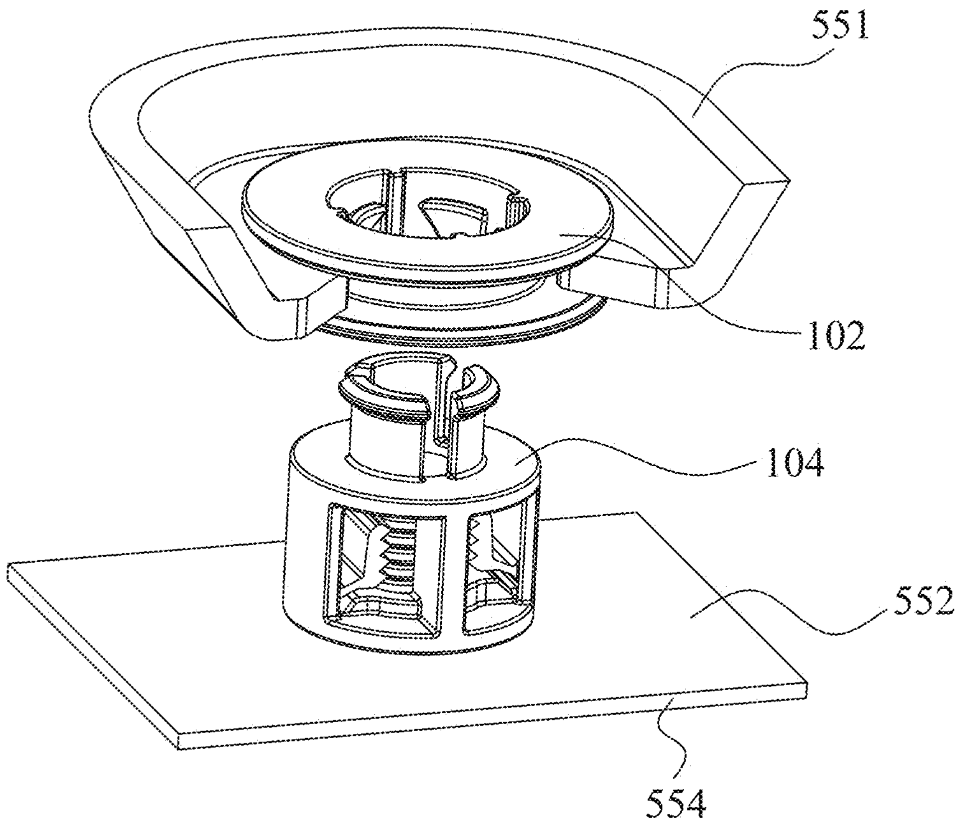
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Fig. 6A

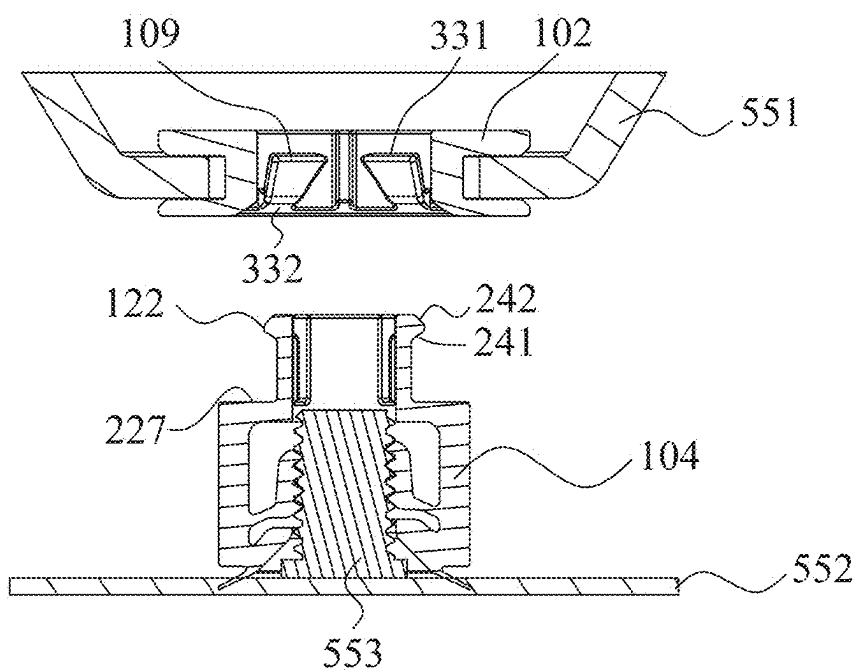
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Fig. 6B

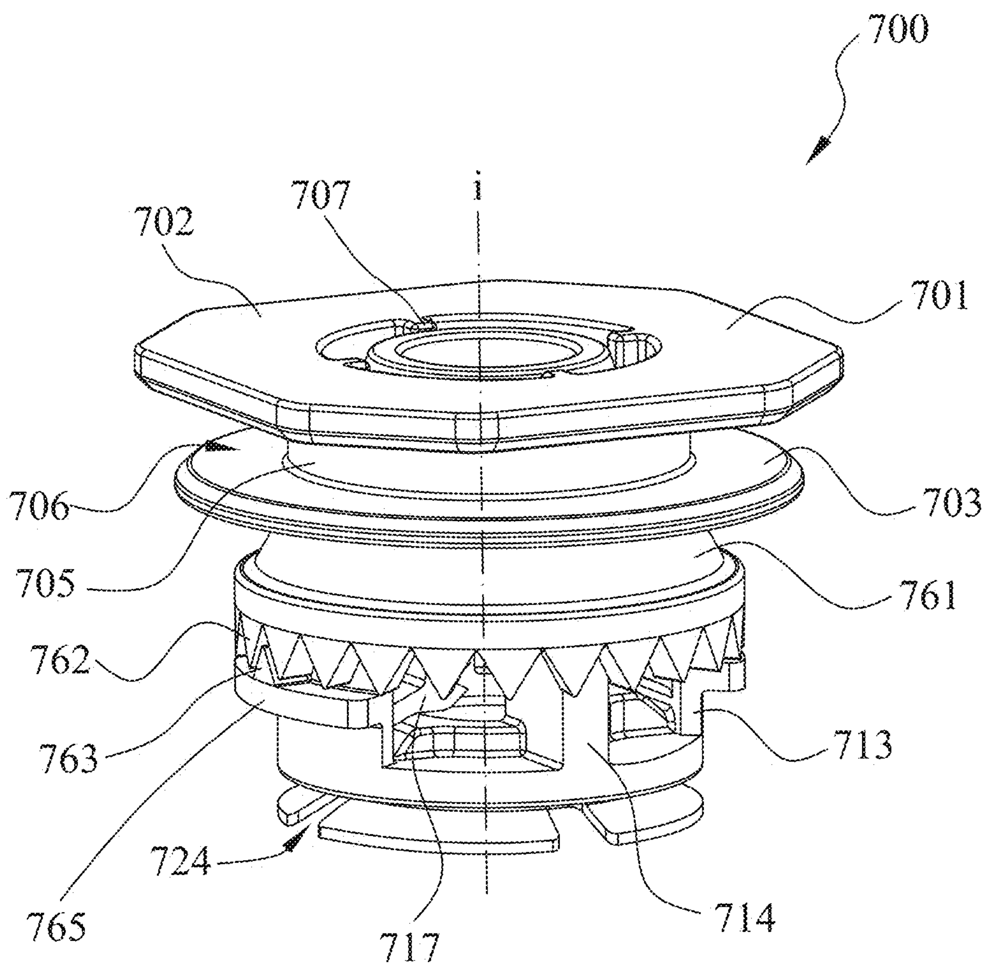

Fig. 7A

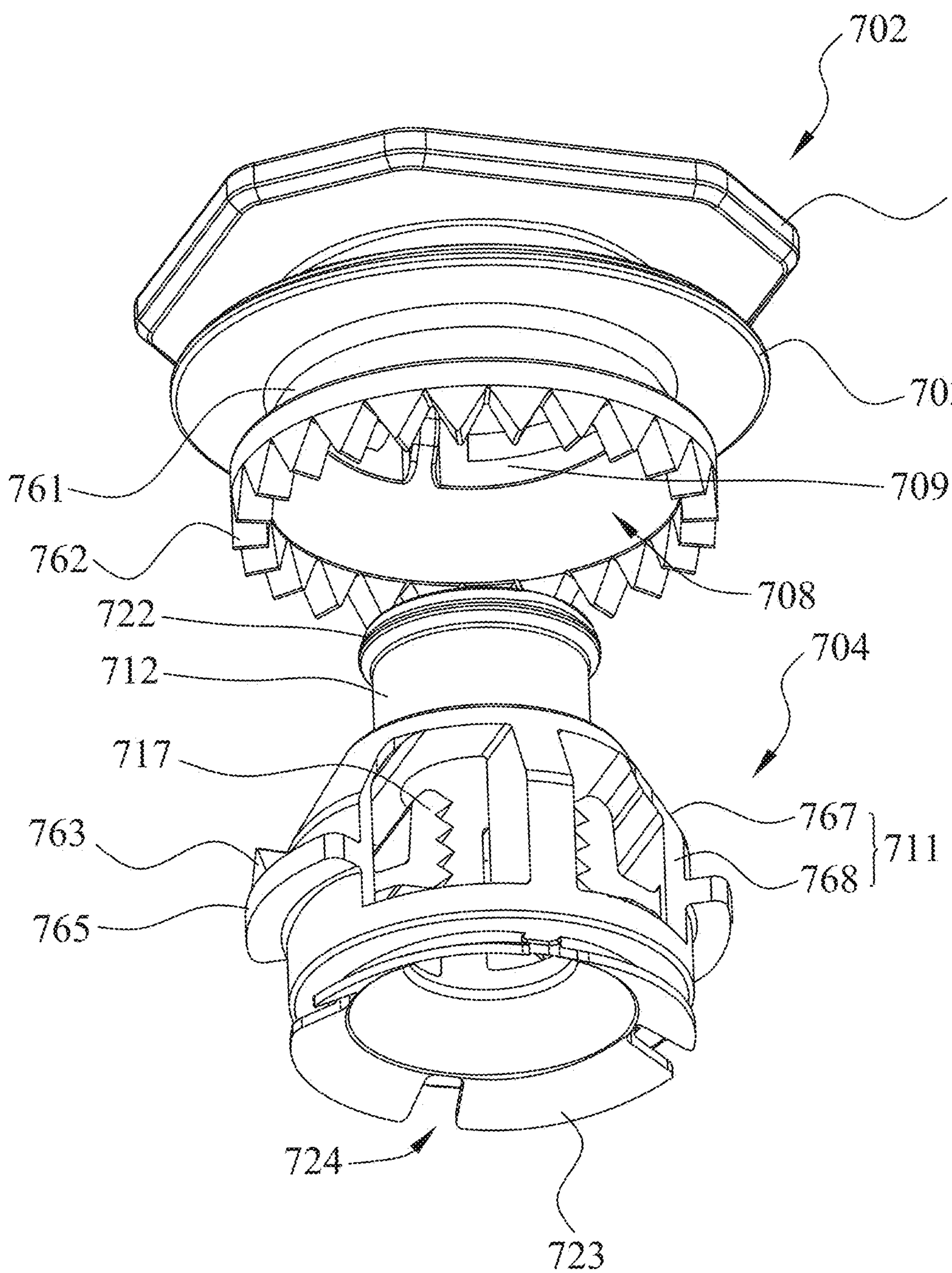
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Fig. 7B

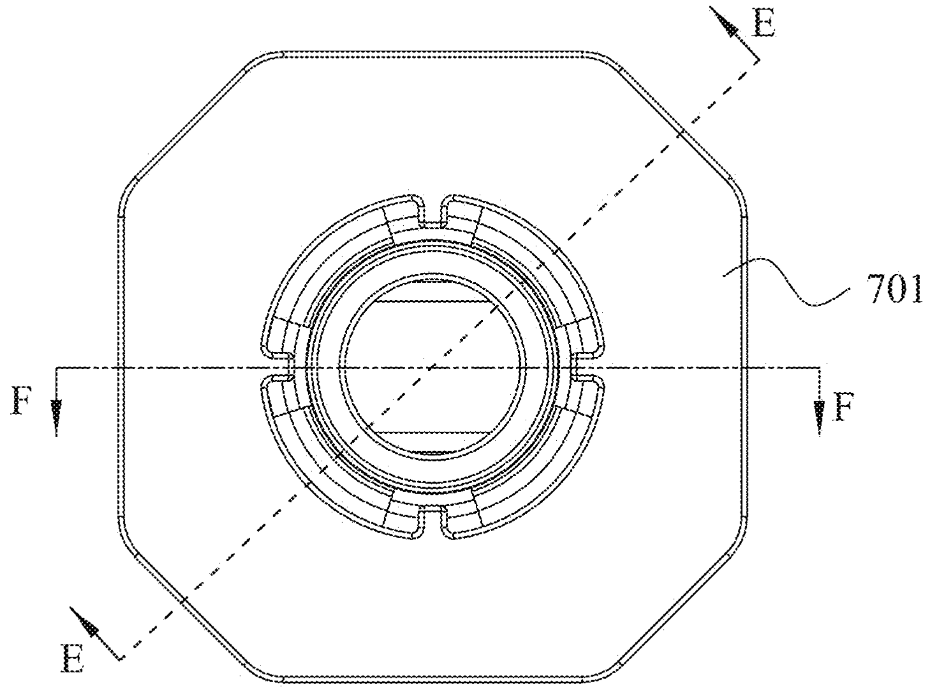


Fig. 7C

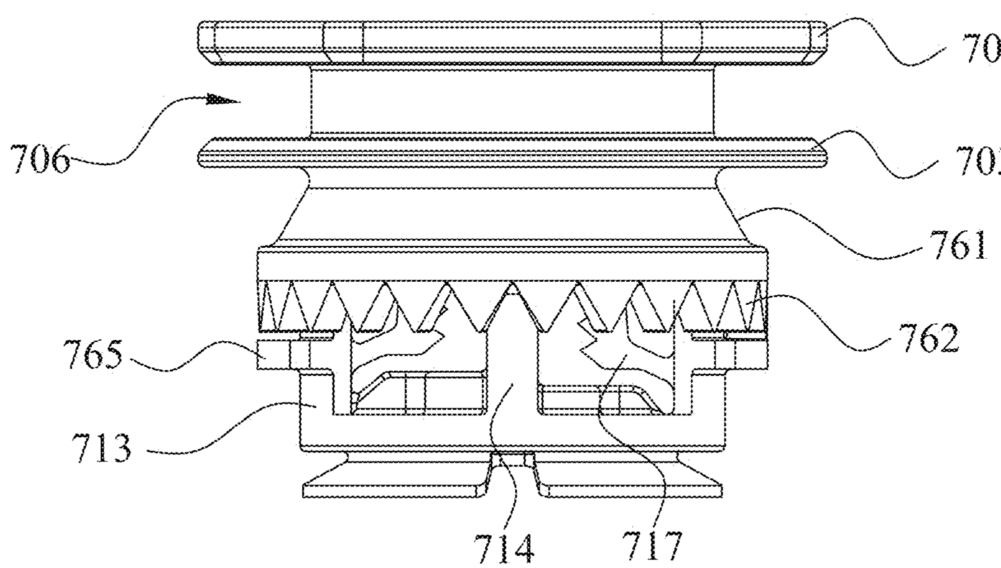
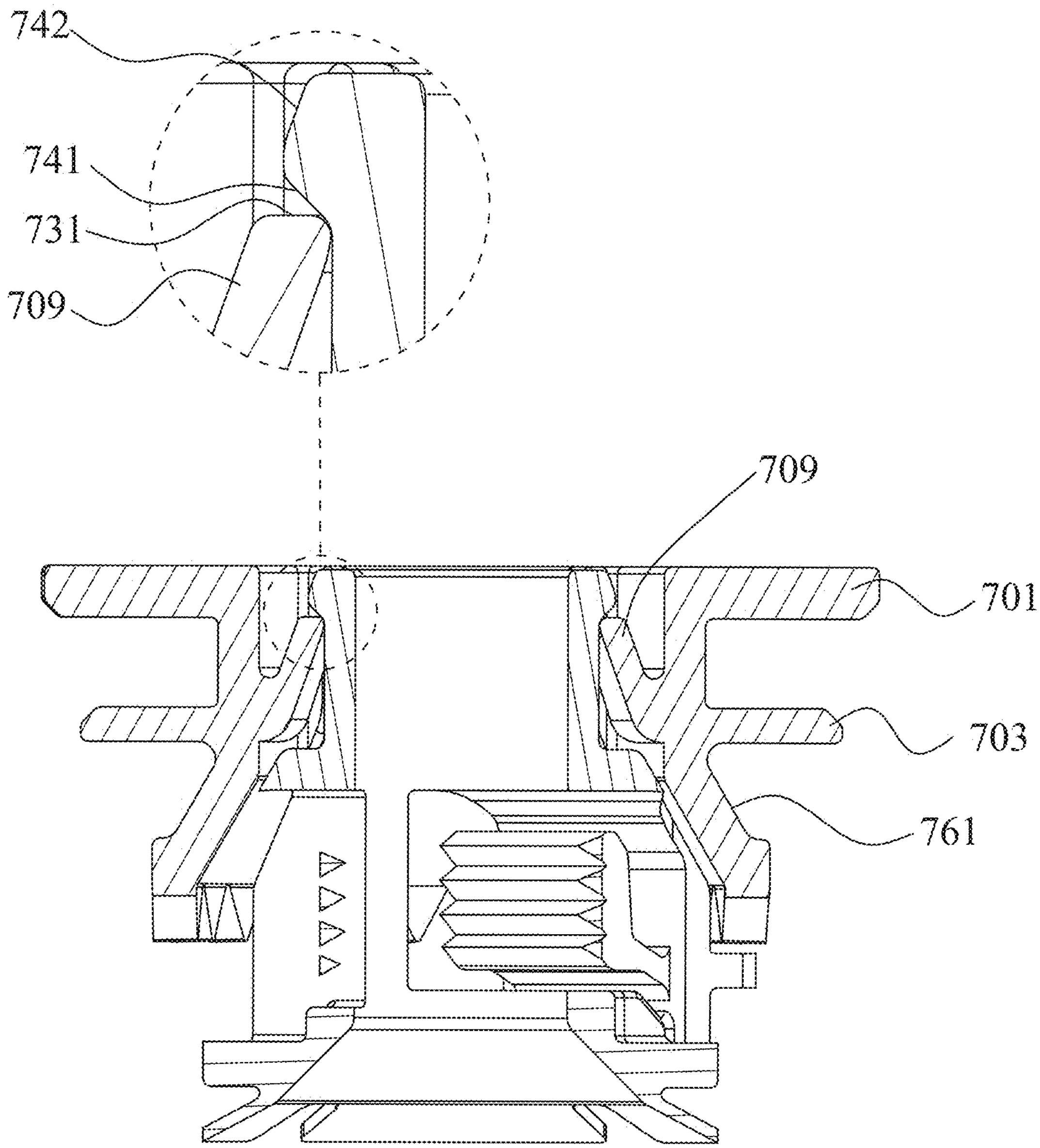


Fig. 7D



E-E

Fig. 7E

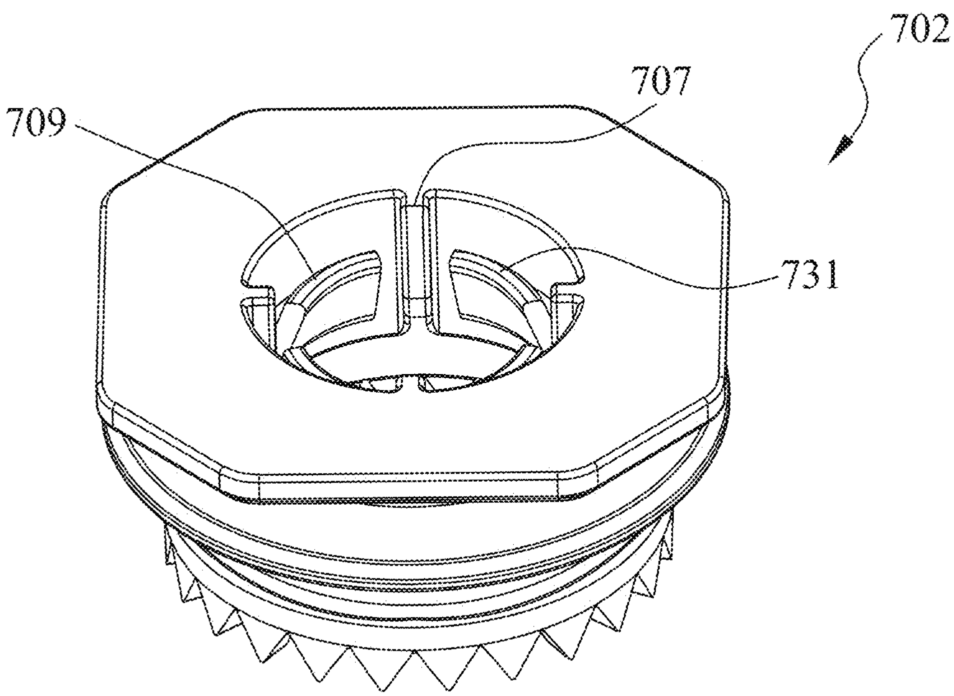


Fig. 8A

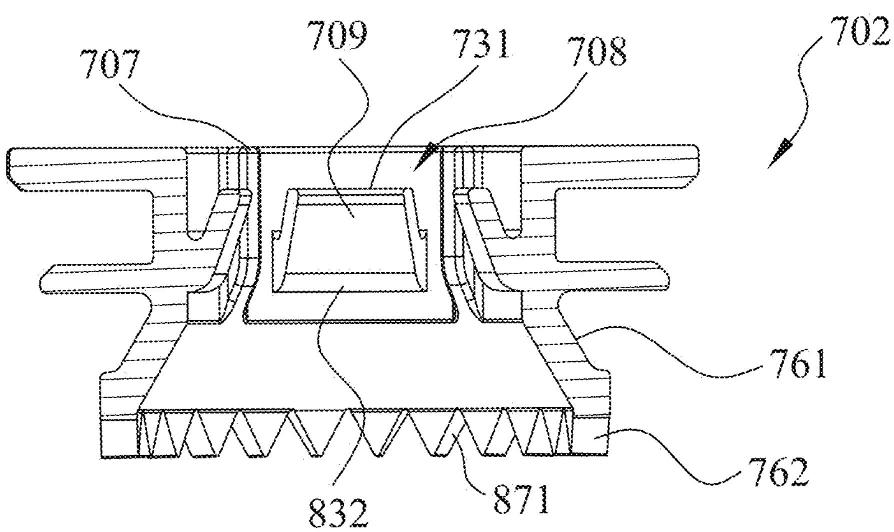


Fig. 8B

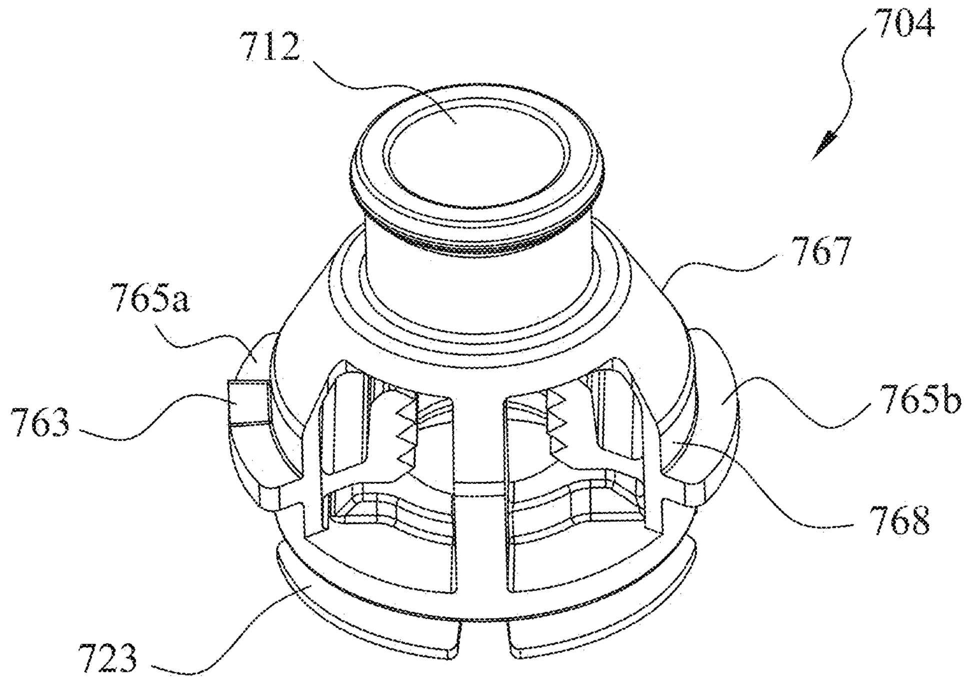


Fig. 9A

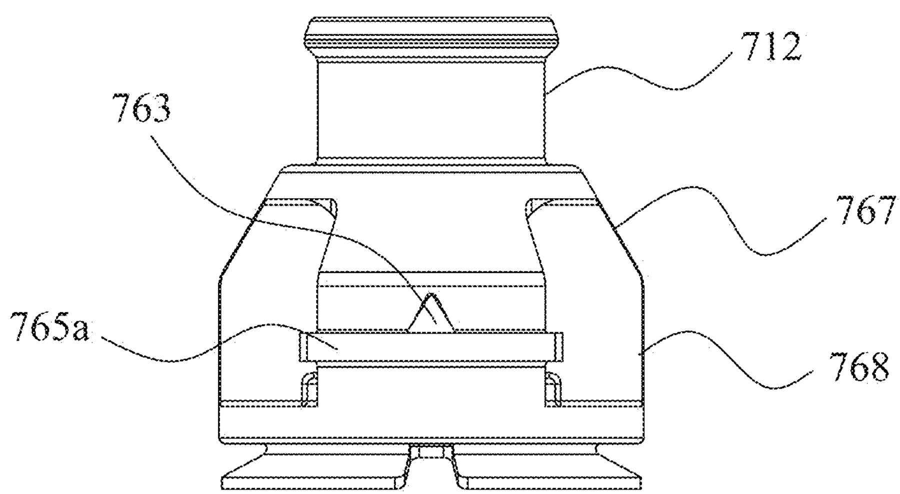


Fig. 9B

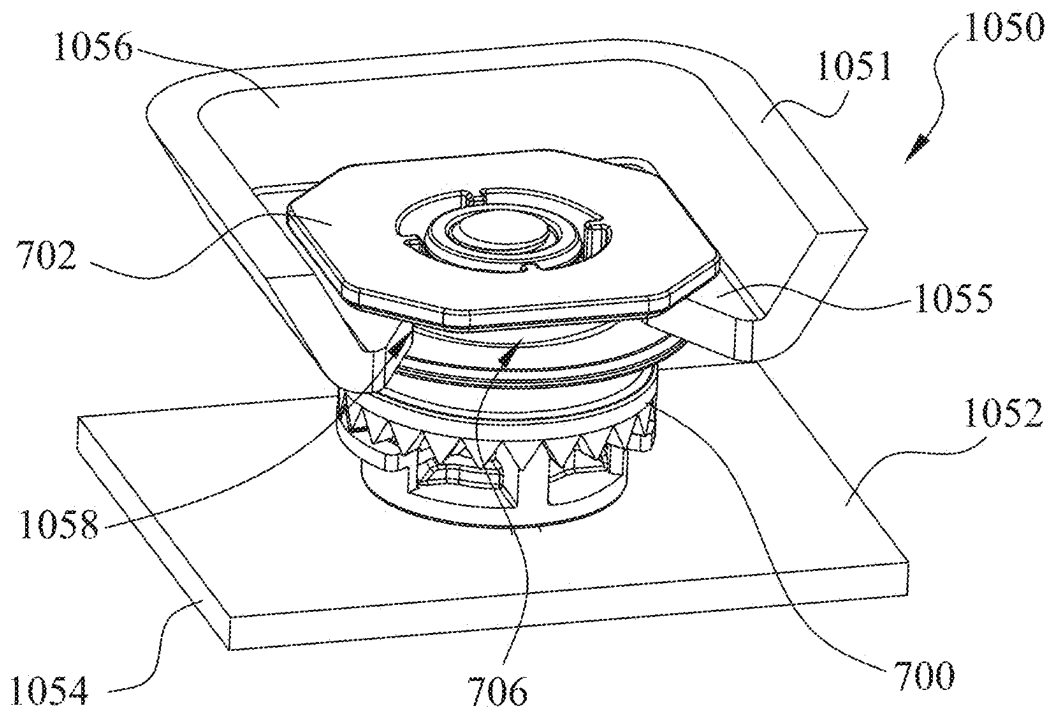


Fig. 10A

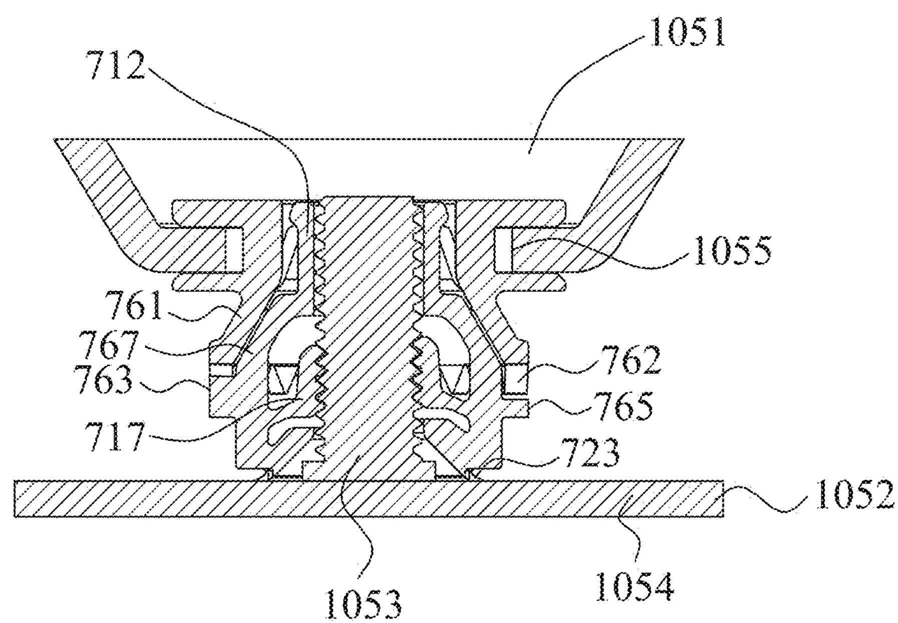


Fig. 10B

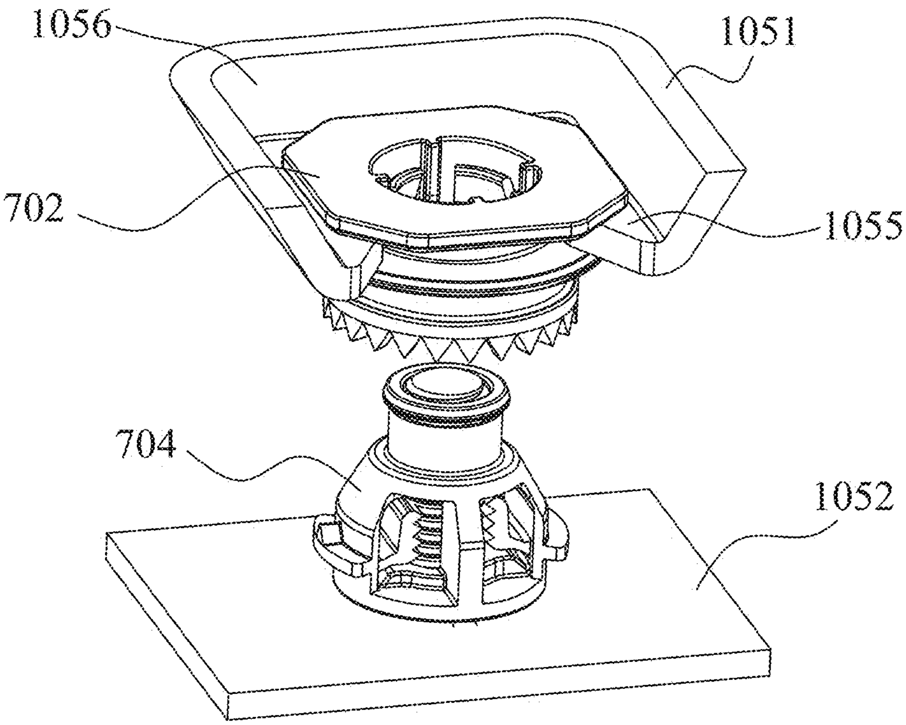


Fig. 11A

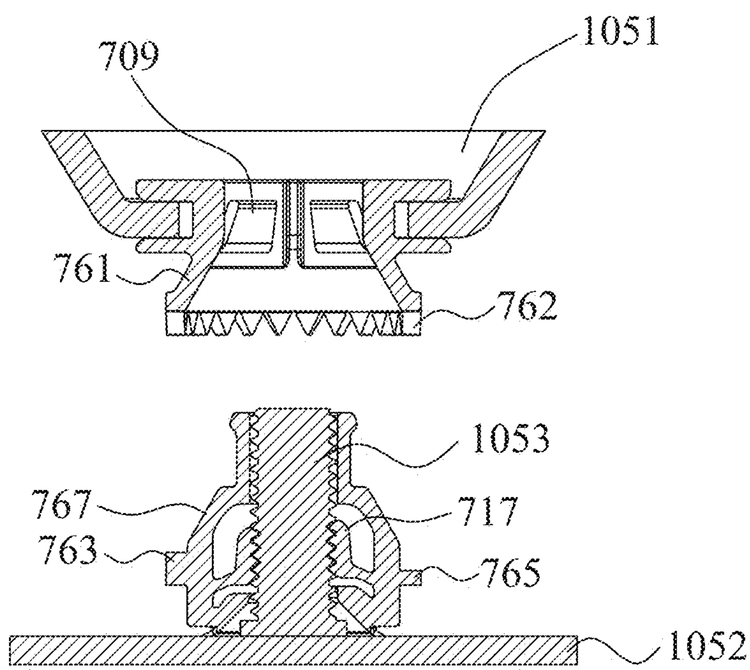


Fig. 11B

FASTENING CLIP AND FASTENING CLIP ASSEMBLY COMPRISING SAME

RELATED APPLICATIONS

[0001] The present application claims priority to the Chinese Patent Application Nos. 202410241795.9, filed Mar. 4, 2024, and 202510196060.3, filed Feb. 21, 2025, each titled "Fastening Clip and Fastening Clip Assembly Comprising Same," the contents of which are hereby incorporated by reference.

BACKGROUND

[0002] In recent years, fastening clips have been developed for fastening a plurality of components to each other. For example, a vehicle includes various components (e.g., metal components or plastic components) that are connected to each other to form a vehicle body, a vehicle door and an interior trim of the vehicle. Some existing fastening clips are usually made of a plastic material. When such existing fastening clips are used to connect two components made of a metal material to each other, it is usually necessary to form holes in the components made of the metal material so as to allow the fastening clip to be inserted into the holes to secure the components and then to connect the two components to each other.

SUMMARY

[0003] The present disclosure relates generally to a fastening clip, substantially as illustrated by and described in connection with at least one of the figures, as set forth more completely in the claims. In one example, the present disclosure relates to a fastening clip, and more particularly to a fastening clip for firmly connecting a plurality of automotive parts together and a fastening clip assembly including the fastening clip.

DESCRIPTION OF THE DRAWINGS

[0004] The foregoing and other objects, features, and advantages of the devices, systems, and methods described herein will be apparent from the following description of particular examples thereof, as illustrated in the accompanying figures; where like or similar reference numbers refer to like or similar structures. The figures are not necessarily to scale, emphasis instead being placed upon illustrating the principles of the devices, systems, and methods described herein.

[0005] FIG. **1**A is a perspective view of a fastening clip at one angle according to one embodiment of the present disclosure.

[0006] FIG. **1**B is an exploded perspective view of the fastening clip shown in FIG. **1**A at another angle.

[0007] FIG. **1**C is a top view of the fastening clip shown in FIG. **1**A.

[0008] FIG. **1**D is a front view of the fastening clip shown in FIG. **1**A.

[0009] FIG. **2**A is a cross-sectional view of the fastening clip shown in FIG. **1**C along line A-A.

[0010] FIG. **2**B is a cross-sectional view of the fastening clip shown in FIG. **1**D along line B-B.

[0011] FIG. **3**A is a perspective view of a first connecter in FIG. **1**A.

[0012] FIG. **3**B is a top view of the first connecter shown in FIG. **3**A.

[0013] FIG. **3**C is a cross-sectional view of the first connecter shown in FIG. **3**B along line C-C.

[0014] FIG. **3**D is a cross-sectional view of the first connecter shown in FIG. **3**B along line D-D.

[0015] FIG. **4**A is a perspective view of a second connecter in FIG. **1**A.

[0016] FIG. **4**B is a top view of the second connecter shown in FIG. **4**A.

[0017] FIG. **5**A is a perspective view of a fastening clip assembly including the fastening clip shown in FIG. **1**A in an assembled state.

[0018] FIG. **5**B is a cross-sectional view of the fastening clip assembly shown in FIG. **5**A.

[0019] FIG. **6**A is a perspective view of the fastening clip assembly including the fastening clip shown in FIG. **1**A in a disassembled state.

[0020] FIG. **6**B is a cross-sectional view of the fastening clip assembly shown in FIG. **6**A.

[0021] FIG. **7**A is a perspective view of a fastening clip according to another embodiment of the present disclosure at an angle.

[0022] FIG. **7**B is an exploded perspective view of the fastening clip shown in FIG. **7**A at another angle.

[0023] FIG. **7**C is a top view of the fastening clip shown in FIG. **7**A.

[0024] FIG. **7**D is a front view of the fastening clip shown in FIG. **7**A.

[0025] FIG. **7**E is a cross-sectional view of the fastening clip shown in FIG. **7**C along line E-E.

[0026] FIG. **8**A is a perspective view of a first connecter in FIG. **7**A.

[0027] FIG. **8**B is a cross-sectional view of the first connecter shown in FIG. **7**C along line E-E.

[0028] FIG. **9**A is a perspective view of a second connecter in FIG. **7**A.

[0029] FIG. **9**B is a left side view of the second connecter in FIG. **9**A.

[0030] FIG. **10**A is a perspective view of a fastening clip assembly including the fastening clip shown in FIG. **7**A in an assembled state.

[0031] FIG. **10**B is a cross-sectional view of the fastening clip assembly shown in FIG. **10**A along line F-F in FIG. **7**C.

[0032] FIG. **11**A is a perspective view of a fastening clip assembly including the fastening clip shown in FIG. **7**A in a disassembled state.

[0033] FIG. **11**B is a cross-sectional view of the fastening clip assembly shown in FIG. **11**A along line F-F in FIG. **7**C.

DETAILED DESCRIPTION

[0034] References to items in the singular should be understood to include items in the plural, and vice versa, unless explicitly stated otherwise or clear from the text. Grammatical conjunctions are intended to express any and all disjunctive and conjunctive combinations of conjoined clauses, sentences, words, and the like, unless otherwise stated or clear from the context. Recitation of ranges of values herein are not intended to be limiting, referring instead individually to any and all values falling within and/or including the range, unless otherwise indicated herein, and each separate value within such a range is incorporated into the specification as if it were individually recited herein. In the following description, it is understood that terms such as "first," "second," "top," "bottom," "side," "front," "back," and the like are words of convenience and are not to be construed as limiting terms. For example, while in some examples a first side is located adjacent or near a second side, the terms "first side" and "second side" do not imply any specific order in which the sides are ordered.

[0035] The terms "about," "approximately," "substantially," or the like, when accompanying a numerical value, are to be construed as indicating a deviation as would be appreciated by one of ordinary skill in the art to operate satisfactorily for an intended purpose. Ranges of values and/or numeric values are provided herein as examples only, and do not constitute a limitation on the scope of the disclosure. The use of any and all examples, or exemplary language ("e.g.," "such as," or the like) provided herein, is intended merely to better illuminate the disclosed examples and does not pose a limitation on the scope of the disclosure. The terms "e.g.," and "for example" set off lists of one or more non-limiting examples, instances, or illustrations. No language in the specification should be construed as indicating any unclaimed element as essential to the practice of the disclosed examples.

[0036] The term "and/or" means any one or more of the items in the list joined by "and/or." As an example, "x and/or y" means any element of the three-element set {(x), (y), (x, y)}. In other words, "x and/or y" means "one or both of x and y". As another example, "x, y, and/or z" means any element of the seven-element set {(x), (y), (z), (x, y), (x, z), (y, z), (x, y, z)}. In other words, "x, y, and/or z" means "one or more of x, y, and z."

[0037] It has been found that if it is necessary to disassemble the two components, it is usually necessary to pull out the fastening clip from the holes of the metal components. This causes the disassembly to occur between the plastic material and the metal material, which can easily result in wear of the plastic material.

[0038] In a first aspect, the present disclosure provides a fastening clip including a first connecter and a second connecter. The first connecter has an axis and a central channel extending along the axis. The first connecter is provided with a groove on an outer peripheral wall. The second connecter includes a main body portion and a sleeve portion. The main body portion is provided with a retaining cavity and at least one retaining arm. The at least one retaining arm is connected to an inner surface of the main body portion at one end and extends from the inner surface of the main body portion toward the retaining cavity. The sleeve portion is connected to the top of the main body portion and is receivable in the central channel of the first connecter. The first connecter and the second connecter are configured to be releasably connected to each other.

[0039] According to the first aspect described above, the first connecter includes at least one snap-fit spring clip. The at least one snap-fit spring clip obliquely extends, from a wall that defines the central channel, toward the central channel. The sleeve portion of the second connecter is releasably connected to the first connecter via the at least one snap-fit spring clip in a snap-fit manner.

[0040] According to the first aspect described above, each of the snap-fit spring clips has a connection end and a free end. One end of each of the snap-fit spring clips is connected to the wall that defines the central channel to form the connection end, and the other end of each of the snap-fit spring clips forms the free end. The at least one snap-fit spring clip is configured to be elastically deformable to allow the sleeve portion of the second connecter to enter the central channel and to be restorable to retain the sleeve portion in the central channel.

[0041] According to the first aspect described above, the sleeve portion includes a step portion formed protruding outwardly from an outer wall of the sleeve portion. The sleeve portion is configured such that when the at least one snap-fit spring clip is restored, the step portion abuts against the free end of the at least one snap-fit spring clip to retain the sleeve portion in the central channel.

[0042] According to the first aspect described above, the step portion has a guided surface at the top of the step portion which extends obliquely outwardly in the top-bottom direction, and/or an inclined bottom end surface at the bottom of the step portion which extends obliquely inwardly in the top-bottom direction. The guided surface is configured to cooperate with the at least one snap-fit spring clip so as to urge the at least one snap-fit spring clip to elastically deform to allow the sleeve portion of the second connecter to enter the central channel. The inclined bottom end surface is configured to cooperate with the at least one snap-fit spring clip so as to abut against the free end of the at least one snap-fit spring clip upon the snap-fit spring clip restores, to retain the sleeve portion in the central channel.

[0043] According to the first aspect described above, the step portion has a flat top end surface which has a smooth edge.

[0044] According to the first aspect described above, the first connecter includes a neck, a top flange and a bottom flange. The central channel is defined in the neck. The top flange is formed extending outwardly from the top of the neck. The bottom flange is formed extending outwardly from the bottom of the neck. The groove is defined to form on an outer side of the neck between the top flange and the bottom flange.

[0045] According to the first aspect described above, the first connecter further includes a barrel portion formed extending downwardly and flaring outwardly from the bottom of the neck. The main body portion of the second connecter includes a first main body formed extending downwardly and flaring outwardly from the bottom of the sleeve portion. The barrel portion and the first main body are matched in shape such that when the first connecter is connected to the second connecter, the first main body abuts against an inner side of the barrel portion.

[0046] According to the first aspect described above, the first connecter is configured to be mounted together with a first component. A first anti-rotation structure is provided on the first connecter and the first component. The first anti-rotation structure includes the top flange in the shape of a polygon with edges and corners.

[0047] According to the first aspect described above, a second anti-rotation structure is provided on the first connecter and the second connecter to prevent the relative rotation between the first connecter and the second connecter.

[0048] According to the first aspect described above, the second anti-rotation structure includes a first toothed portion disposed at a bottom end of the barrel portion of the first connecter, and a second toothed portion connected to an outer side of the main body portion of the second connecter. The first toothed portion and the second toothed portion mesh and cooperate with each other to prevent the relative rotation between the first connecter and the second connecter.

[0049] According to the first aspect described above, the first toothed portion and the second toothed portion are triangular teeth.

[0050] According to the first aspect described above, the first toothed portion includes a ring of teeth arranged in a circumferential direction, and the second toothed portion includes a single tooth.

[0051] According to the first aspect described above, the second connecter further includes at least one support boss disposed on the outer side of the main body portion and configured to be capable of at least partially supporting the first toothed portion of the first connecter when the first connecter is connected to the second connecter.

[0052] According to the first aspect described above, the sleeve portion includes at least one elastic leg configured to be releasably connected to the at least one snap-fit spring clip in a snap-fit manner.

[0053] According to the first aspect described above, the fastening clip is made of a plastic material.

[0054] In a second aspect, the present disclosure provides a fastening clip assembly, including a first component, a second component and a fastening clip according to the first aspect. The second component includes a body and a pin which are fixedly connected to each other. The first component is received by the groove of the first connecter, and the pin is capable of being cooperatively connected to the retaining arm of the second connecter of the fastening clip to securely connect the first component to the second component.

[0055] According to the second aspect described above, the first component includes a flat plate portion and a ring-shaped wall disposed around a peripheral edge of the flat plate portion. The flat plate portion is engaged within the groove of the first connecter. A first anti-rotation structure is provided on the first connecter and the first component. The first anti-rotation structure including the top flange in the shape of a polygon with edges and corners and a ring-shaped wall matching the shape of the top flange, to prevent the relative rotation between the first connecter and the first component.

[0056] According to the first aspect described above, one of the first component and the second component is made of a metal material and the other is made of a plastic material.

[0057] The present disclosure provides a fastening clip including a first connecter and a second connecter which are releasably connected to each other. When the fastening clip connects two components (especially for fixing vehicle parts), the first connecter and the second connecter are firmly connected to the two components, respectively. Moreover, the two components are separated from each other by disassembling the first connecter and the second connecter of the fastening clip of the present disclosure, so that it is possible to avoid wear between the components made of a metal material and the fastening clip made of a plastic material. The fastening clip of the present disclosure is particularly suitable for connecting two components with at least one being made of a metal material.

[0058] Other objectives and advantages of the present disclosure will be apparent from the following description of the present disclosure with reference to the drawings, which can contribute to a comprehensive understanding of the present disclosure.

[0059] FIGS. **1**A-**1**D show the basic structure of a fastening clip **100** according to one embodiment of the present disclosure. FIG. **1**A shows a perspective view of the fastening clip **100** from a top view. FIG. **1**B shows a perspective view of the fastening clip **100** from a different bottom view. FIG. **1**C shows a top view of the fastening clip **100**. FIG. **1**D shows a front view of the fastening clip **100**. As shown in FIGS. **1**A-**1**D, the fastening clip **100** includes a first connecter **102** and a second connecter **104**. The first connecter **102** and the second connecter **104** are releasably connected together. The first connecter **102** is configured to connect a first component **551** to be securely connected, and a second connecter **104** is configured to connect a second component **552** to be securely connected (see FIGS. **5**A and **6**A). Thus, by disassembling and assembling the first connecter **102** and the second connecter **104**, the first component **551** and the second component **552** can be releasably connected together. In this embodiment, the second connecter **104** and the first connecter **102** have coaxial shapes, and the second connecter **104** is inserted axially into the first connecter **102** and releasably connected to the first connecter **102**. Moreover, in this embodiment, the first connecter **102** and the second connecter **104** are releasably connected together in a snap-fit manner. It can be understood by those skilled in the art that the first connecter **102** and the second connecter **104** may be connected together in any other releasable manner.

[0060] In this embodiment, the first connecter **102** is substantially in the shape of a cylinder having an axis i. The first connecter **102** includes a neck **105** and a central channel **108**. The neck **105** is in the shape of a circular ring with the central channel **108** running through the center of the neck **105**. The central channel **108** extends along the axis i.

[0061] The first connecter **102** further includes at least one snap-fit spring clip **109**. One end of each snap-fit spring clip **109** is connected to a wall that defines the central channel **108** to form a connection end. That is, the at least one snap-fit spring clip **109** is connected to an inner surface of the neck **105**. The at least one snap-fit spring clip **109** extends obliquely into the central channel **108**. The other end of each snap-fit spring clip **109** forms a free end. In this way, each snap-fit spring clip **109** have a certain elasticity, and a distance between the free end of each snap-fit spring clip **109** and the axis i is less than a distance between its connection end and the axis i. The first connecter **102** further includes at least one guide strip **107** also disposed on the wall that defines the central channel **108**. Put it differently, the at least one guide strip **107** is disposed on the inner surface of the neck **105**. The at least one guide strip **107** extends parallel to the axis i. In this embodiment, the first connecter **102** includes four snap-fit spring clips **109** and four guide strips **107**. The four snap-fit spring clips **109** are arranged at intervals circumferentially around the axis i, and the four guide strips **107** are each disposed between two adjacent snap-fit spring clips **109**. The snap-fit spring clips **109** are configured to connect the second connecter **104** in a snap-fit manner, and the central channel **108** is configured for receiving at least part of the second connecter **104**. The more specific structures of the snap-fit spring clip **109** and the guide strip **107** will be described in detail with reference to FIGS. **3**A-**3**D.

[0062] The first connecter **102** further includes a top flange **101**, a bottom flange **103** and a groove **106**. The top flange **101** is formed extending outwardly from the top of the neck **105**, for example, extending horizontally in a radial direction. The bottom flange **103** is formed extending outwardly from the bottom of the neck **105**, for example, extending horizontally in a radial direction. In this way, the first connecter **102** has the groove **106** between the top flange **101** and the bottom flange **103**. The groove **106** is configured to receive a plate to be securely connected, such as a flat plate portion **555** of the first component **551** (see FIGS. **5**A and **5**B). In this embodiment, the groove **106** is a circular ring-shaped groove. It can be understood by those skilled in the art that the groove **106** of the first connecter **102** may also be provided as a groove in the shape of a square ring, a polygonal ring or any other shapes, to receive the first component of another shape to be securely connected. Moreover, although in this embodiment, the top flange **101** and the bottom flange **103** are circular ring-shaped flanges, in other embodiments, the top flange **101** or the bottom flange **103** may be configured to be flanges having an outer peripheral wall in the shape of a square or other polygonal shapes.

[0063] The second connecter **104** includes a main body portion **111**, a sleeve portion **112** and a flared mouth portion **123** having a common axis i. The sleeve portion **112** is connected to the top of the main body portion **111**, and the flared mouth portion **123** is connected to the bottom of the main body portion **111**. The sleeve portion **112** is in the shape of a cylinder for insertion into the central channel **108** of the first connecter **102** and the snap-fit connection with the snap-fit spring clips **109** in the central channel **108**. The top end of the sleeve portion **112** has step portions **122** protruding outwardly. The distance between the distal ends of the step portions **122** and the axis i is set to be less than the distance between the connection end of the snap-fit spring clips **109** and the axis i, but greater than the distance between the free ends of the snap-fit spring clips **109** and the axis i. When the first connecter **102** and the second connecter **104** are assembled together, the step portions **122** abut against the top ends of the snap-fit spring clips **109** of the first connecter **102** to prevent the downward movement of the second connecter **104** relative to the first connecter **102**. In some embodiments, the sleeve portion **112** includes at least one elastic leg **121**. The distance between an outer surface of the elastic leg **121** and the axis i is substantially equal to the distance between the free ends of the snap-fit spring clips **109** and the axis i such that when the first connecter **102** and the second connecter **104** are assembled together, the outer surface of the elastic leg **121** is close to the tops of the snap-fit spring clips **109** to prevent the excessive radial displacement of the sleeve portion **112** relative to the first connecter **102**. In this embodiment, the sleeve portion **112** include three elastic legs **121** arranged at intervals circumferentially around the axis i. By providing the elastic legs **121**, the sleeve portion **112** also has a certain elasticity, thereby controlling an insertion force and a retaining force between the first connecter **102** and the second connecter **104**. It can be understood by those skilled in the art that in other embodiments, it is also possible that the sleeve portion **112** includes no elastic legs, but instead has a complete cylindrical shape. In such embodiments, only the snap-fit spring clips **109** of the first connecter **102** provide elasticity. Thus, the insertion force and the retaining force between the first connecter **102** and the second connecter **104** can be controlled by simply designing the shape of the snap-fit spring clips **109** as needed. The more specific structure of the sleeve portion **112** will be described in detail with reference to FIGS. **2**A and **2**B.

[0064] In this embodiment, the main body portion **111** is substantially in the shape of a cylinder, including a pair of connection beams **113** and a pair of support beams **114** connected between the top and the bottom of the main body portion **111**. The pair of connection beams **113** are symmetrically disposed with respect to the axis i, and the pair of support beams **114** are also symmetrically disposed with respect to the axis i. The interior of the main body portion **111** defines a hollow retaining cavity **118**. The main body portion **111** further includes a pair of retaining arms **117**. Each of the pair of retaining arms **117** extends at one end from an inner surface of a respective one of the pair of connection beams **113** to the retaining cavity **118**. In some embodiments, each retaining arm **117** is connected to the respective connection beam **113** via an integrally formed pivot portion **115**. Each retaining arm **117** includes thread engaging cogs **119** arranged parallel to the axis i. The thread engaging cogs **119** can engage with a pin **553** having an external thread (see FIGS. **5**A and **5**B) to retain the pin **553** in the retaining cavity **118** by means of the retaining arms **117**. In this embodiment, the pair of retaining arms **117** includes a first retaining arm **117***a* and a second retaining arm **117***b*. The two retaining arms **117** are disposed substantially symmetrically in the circumferential direction and their thread engaging cogs **119** are located at different height levels, such that the pair of retaining arms **117** can engage with the external thread of a pin **553** having a greater length along the axial direction. It can be understood by those skilled in the art that the number of retaining arms **117** may be set to be more or less, such that there is a greater or lesser insertion force and retaining force between the main body portion **111** and the pin **553**. Moreover, it can be understood by those skilled in the art that in some embodiments, the retaining arms may be configured to have other shapes and structures depending on the different structures of the inserted pin.

[0065] The flared mouth portion **123** is substantially in the shape of a bell mouth, which is connected to the bottom of the main body portion **111**. The flared mouth portion **123** includes at least one notch **124**. The notch **124** can facilitate the flared mouth portion **123** opening and deformation to a certain degree. In this embodiment, the flared mouth portion **123** is provided with two notches **124** symmetrically disposed at the bottom of the flared mouth portion **123**. In this way, the flared mouth portion **123** can abut against a body **554** of the second component **552** (see FIG. **5**B) to be connected. This provides upward force to the pin **553** of the second component **552**, pressing the external thread of the pin **553** against the thread engaging cogs **119** of the retaining arms **117**. The flared mouth portion **123** can match and accommodate the greater-diameter head of the pin **553** when the pin **553** is being inserted into the retaining cavity **118**. It can be understood by those skilled in the art that the bottom edge of the flared mouth portion **123** may be configured to have a suitable outer contour to match different shapes of the body **554** of the second component **552**. It can be understood by those skilled in the art that the flared mouth portion is not essential for the present disclosure. In some embodiments, the second connecter may alternatively include no flared mouth portion. In some embodiments, the second connecter may include the flared mouth portion, but not include the notch.

[0066] It should be noted that the present disclosure is not limited to the structure of the main body portion described in this embodiment. The shape and structure of the main body portion can also be configured according to the different shapes and engagement requirements of the second component.

[0067] In the present disclosure, the first connecter **102** and the second connecter **104** of the fastening clip **100** may both be made of a plastic material. The first connecter **102** and the second connecter **104** that are made of a plastic material may have a good elasticity to facilitate repeated disassembly and assembly.

[0068] FIGS. **2**A and **2**B are used to more clearly show the internal structure of the fastening clip **100**. FIG. **2**A is a cross-sectional view of the fastening clip **100** along line A-A, and FIG. **2**B is a cross-sectional view of the fastening clip **100** along line B-B. As shown in FIGS. **2**A and **2**B, the sleeve portion **112** of the second connecter **104** has a hollow sleeve channel **228**. In this embodiment, the elastic legs **121** can be elastically deformed to a certain degree into the sleeve channel **228** when the outer surfaces of the elastic legs **121** of the sleeve portion **112** are pressed.

[0069] The bottoms of the step portions **122** of the sleeve portion **112** has an inclined bottom end surface **241**. The inclined bottom end surfaces **241** are configured to cooperate with the tops of the snap-fit spring clips **109**, so as to prevent the downward movement of the second connecter **104** relative to the first connecter **102** when the first connecter **102** and the second connecter **104** are assembled together. In this embodiment, the inclined bottom end surfaces **241** are inclined from bottom to top in a direction gradually away from the axis i. By providing the inclined bottom end surfaces **241** at the bottoms of the step portions **122**, when an operator pulls the second connecter **104** out of the first connecter **102**, a component force that urges the deformation of the snap-fit spring clips **109** can be generated at the contact positions between the inclined bottom end surfaces **241** of the step portions **122** and the snap-fit spring clips **109**, thereby facilitating disassembly of the first connecter **102** from the second connecter **104**. The tops of the step portions **122** have a guided surface **242**. The guided surfaces **242** are configured to cooperate with the bottoms of the snap-fit spring clips **109**. In this embodiment, the guided surfaces **242** are inclined from bottom to top in a direction gradually close to the axis i. By providing the guided surfaces **242** at the tops of the step portions **122**, when the operator inserts the second connecter **104** into the first connecter **102**, a component force that urges the deformation of the snap-fit spring clips **109** can be generated at the contact positions between the guided surfaces **242** of the step portions **122** and the snap-fit spring clips **109**, thereby facilitating assembly of the first connecter **102** and the second connecter **104**.

[0070] The top of the main body portion **111** of the second connecter **104** has a shoulder **227**. The shoulder **227** is formed extending substantially horizontally from a connection portion of the bottom of the sleeve portion **112** and the main body portion **111**. When the first connecter **102** and the second connecter **104** are assembled together, the shoulder **227** abuts against a lower surface of the bottom flange **103** of the first connecter **102** to prevent the upward movement of the second connecter **104** relative to the first connecter **102**. When the first connecter **102** and the second connecter **104** are assembled together, the step portions **122** and the shoulder **227** together ensure that the first connecter **102** and the second connecter **104** cannot be displaced in the axial direction. Moreover, by disposing the outer diameter of the sleeve portion **112** beneath the step portions **122**, the radial and circumferential displacements of the first connecter **102** and the second connecter **104** can be prevented. In this way, the first connecter **102** and the second connecter **104** can keep the snap-fit connected state when they are assembled together.

[0071] The pair of connection beams **113** and the pair of support beams **114** of the main body portion **111** are spaced apart from each other in the circumferential direction of the main body portion **111**. The connection beams **113** have a greater width (i.e., the dimension in the circumferential direction) but a smaller thickness (i.e., the dimension in the radial direction) than the support beams **114**. This is because the connection beams **113** are configured to be provided with the retaining arms **117***a* and **117***b* on the inner walls, which requires a space to be reserved in the thickness direction, and because a greater width is required in order to increase the strength of the retaining arms **117***a* and **117***b*. The support beams **114** have a greater thickness and a smaller width because the support beams **114** are used to provide strength to the main body portion **111** and a space for machining the retaining arms needs to be reserved.

[0072] FIGS. **3**A-**3**D show a more specific structure of the first connecter **102**. FIG. **3**A shows a perspective view of the first connecter **102**, FIG. **3**B shows a top view of the first connecter **102**, FIG. **3**C shows a cross-sectional view of the first connecter **102** along line C-C, and FIG. **3**D shows a cross-sectional view of the first connecter **102** along line D-D. As shown in FIGS. **3**A-**3**D, an inner wall of the neck **105** of the first connecter **102** defines the central channel **108**. The four snap-fit spring clips **109** are circumferentially disposed at intervals on the inner wall of the neck **105**.

[0073] In this embodiment, the sleeve portion **112** of the second connecter **104** is inserted into the central channel **108** in the bottom-top direction. The snap-fit spring clips **109** are thus connected at the bottom to the inner wall of the neck **105** and extend from bottom to top obliquely toward the axis i, so that the tops of the snap-fit spring clips **109** form the free ends closest to the axis i. When the snap-fit spring clips **109** are subjected to an outward push force, the snap-fit spring clips **109** deform with pivoting about their bottoms so that the tops of the snap-fit spring clips **109** move away from the axis i, thereby creating an insertion channel and an exit channel for the sleeve portion **112**. In this embodiment, the bottom of each snap-fit spring clip **109** has a guide surface **332** extending from bottom to top obliquely toward the axis i. The guide surfaces **332** cooperate with the guided surfaces **242** on the tops of the step portions **122** of the sleeve portion **112** to guide the insertion of the sleeve portion **112** into the central channel **108**. Moreover, in this embodiment, the top of each snap-fit spring clip **109** has an inclined top end surface **331** extending from bottom to top obliquely away from the axis i. The inclined top end surfaces **331** are in form-fit with the inclined bottom end surfaces **241** of the step portions **122** of the sleeve portion **112**, such that when the second connecter **104** is pulled out of the central channel **108**, a component force which pushes the snap-fit spring clips **109** outward can be also generated at the tops of the snap-fit spring clips **109**, to deform the snap-fit spring clips **109**, thereby enabling the sleeve portion **112** to be pulled out of the central channel **108**. In this embodiment, the width of each snap-fit spring clip **109** in the circumferential direction gradually decreases from the top to the bottom, which reduces the force required to deform the snap-fit spring clips **109**. In other embodiments, the top ends of the snap-fit spring clips **109** may not be in the shape of an inclined surface, but a component force that urges the deformation of the snap-fit spring clips **109** can also be generated when the second connecter **104** is being pulled out.

[0074] In some practical assembly processes, although the first connecter **102** and the second connecter **104** are coaxially arranged, it is not easy to ensure that they are inserted perfectly axially and coaxially during their assembly. The provision of the axially-extending guide strips **107** between the four snap-fit spring clips **109** enables that the guide strips **107** can guide the sleeve portion **112** to be inserted into or pulled out of the central channel **108** even if there is an assembly error between the first connecter **102** and the second connecter **104**. Moreover, the guide strips **107** can bear a shear force caused by the misalignment of at least some of the snap-fit spring clips **109** and the corresponding spaces between the elastic legs **121** during the assembly, thereby protecting the snap-fit spring clips **109** from damage.

[0075] FIGS. **4**A and **4**B show a more specific structure of the second connecter **104**. FIG. **4**A is a perspective view of the second connecter **104**, and FIG. **4**B is a top view of the second connecter **104**. As shown in FIGS. **4**A and **4**B, in this embodiment, the sleeve portion **112** includes three elastic legs **121** arranged at intervals, each elastic leg **121** being respectively provided with a step portion **122** at the top. When the step portions **122** apply a push force to the snap-fit spring clips **109** to push the snap-fit spring clips **109**, the elastic legs **121** may also deflect to a certain extent toward the axis i, thereby reducing the force causing the deformation of the snap-fit spring clips **109**. In some embodiments, it is also possible that no elastic leg **121** is provided, so that the sleeve portion **112** is inserted into and pulled out of the central channel **108** in its entirety.

[0076] Moreover, in the present disclosure, the number of the snap-fit spring clips **109** and the number of the elastic legs **121** are set to be different. This can reduce the difficulty of assembling and disassembling the first connecter **102** and the second connecter **104**, so that the operator does not need to align all the snap-fit spring clips **109** with the spaces between the elastic legs **121**.

[0077] In this way, during insertion of the sleeve portion **112** into the central channel **108** along the axis i of the first connecter **102**, the step portions **122** push away the snap-fit spring clips **109**, forcing the snap-fit spring clips **109** to be elastically deformed toward the neck **105**, to create the insertion channel and the exit channel for the sleeve portion **112**. After the sleeve portion **112** is inserted into the central channel **108** of the first connecter **102**, the step portions **122** pass over the top end of the snap-fit spring clips **109**. This causes that the snap-fit spring clips **109** are no longer pressed by the step portions **122** and return to their original shape. Moreover, the bottoms of the step portions **122** abut against the tops of the snap-fit spring clips **109**. The first connecter **102** and the second connecter **104** are thus connected to each other in a snap-fit manner. In a reverse process, the first connecter **102** and the second connecter **104** can be disassembled by pulling the sleeve portion **112** out of the second connecter **104** from the central channel **108** of the first connecter **102** along the axis i.

[0078] FIGS. **5**A and **5**B show the structure of a fastening clip assembly **550** including the fastening clip **100** in an assembled state. FIG. **5**A shows a perspective view of the fastening clip assembly **550**, and FIG. **5**B shows a cross-sectional view of the fastening clip assembly **550** taken along line A-A in FIG. **1**C. As shown in FIGS. **5**A and **5**B, the fastening clip assembly **550** includes a first component **551**, a second component **552**, and the fastening clip **100**. The first component **551** and the first connecter **102** of the fastening clip **100** are connected together. In this embodiment, the first component **551** includes a substantially circular flat plate portion **555** and a ring-shaped wall **556**. The flat plate portion **555** is provided with an opening **558** at the front side. The ring-shaped wall **556** is disposed around a peripheral edge of the flat plate portion **555**. The flat plate portion **555** is engaged in the groove **106** of the first connecter **102** through the opening **558**, and is thus received by the groove **106**. The second component **552** and the second connecter **104** of the fastening clip **100** are connected together. In this embodiment, the second component **552** includes a body **554** and a pin **553**. The body **554** and the pin **553** are fixedly connected together. The pin **553** can be inserted into the retaining cavity **118** of the second connecter **104** in the bottom top direction, and engages with the retaining arms **117**. Moreover, the flared mouth portion **123** abuts against an upper surface of the body **554**. In this way, the first component **551** and the second component **552** can be securely connected to each other. As an example, the pin **553** is a weld stud having an external thread that mates with the thread engaging cogs **119** of the retaining arms **117**. The pin **553** engages with the thread engaging cogs **119** of the retaining arms **117** by means of the external thread, so that the pin **553** is firmly connected to the second connecter **104**.

[0079] In the present disclosure, the second component **552** is made of a metal material such as sheet metal of a vehicle. The body **554** is fixedly connected with the pin **553** by welding. The first component **551** is another plastic or metal component of the vehicle. Moreover, the fastening clip **100** is made of a plastic material with a certain elasticity.

[0080] It can be understood by those skilled in the art that the shapes of the first component **551** and the second component **552** are shown schematically. In fact, it is sufficient as long as the first component **551** includes the flat plate portion **555** and the second component **552** includes the pin **553**.

[0081] FIGS. **6**A and **6**B show the structure of the fastening clip assembly **550** including the fastening clip **100** in a disassembled state. FIG. **6**A shows a perspective view of the fastening clip assembly **550**, and FIG. **6**B shows a cross-sectional view of the fastening clip assembly **550** taken along line A-A in FIG. **1**C. As shown in FIGS. **6**A and **6**B, when the operator pulls the first component **551** upwardly or pulls out the second component **552** downwardly, the fastening clip assembly **550** is changed from the assembled state to the disassembled state. At this moment, the inclined bottom end surfaces **241** of the step portions **122** of the second connecter **104** comes into contact with the inclined top end surfaces **331** of the snap-fit spring clips **109** and a component force that urges the deformation of the snap-fit spring clips **109** is generated, so that the snap-fit spring clips **109** are deformed and the sleeve portion **112** can be pulled out of the central channel **108**. Moreover, at this moment, the first connecter **102** and the first component **551** remain connected together and the second connecter **104** and the second component **552** remain connected together, but the first connecter **102** is disassembled from the second connecter **104** so that the first component **551** is disconnected from the second component **552**.

[0082] By setting the retaining force between the first component **551** and the first connecter **102**, the retaining force between the first connecter **102** and the second connecter **104**, and the retaining force between the second connecter **104** and the second component **552** within proper ranges, it is possible to make the disassembly of the first connecter **102** and the second connecter **104** easier than the disassembly of the other components, thereby avoiding wear between the fastening clip made of plastic and the first and/or second component made of metal. As a specific example, the retaining force between the first component **551** and the first connecter **102** may be up to 600 N and the retaining force between the second connecter **104** and the second component **552** may be up to 600 N, while the retaining force between the first connecter **102** and the second connecter **104** is only 250 N.

[0083] When the operator pushes the first component **551** or the second component **552**, the fastening clip assembly **550** can be changed from the disassembled state to the assembled state. At this moment, the guided surfaces **242** of the step portions **122** of the second connecter **104** come into contact with the guide surfaces **332** of the snap-fit spring clips **109** and a component force that urges the deformation of the snap-fit spring clips **109** is generated, so that the snap-fit spring clips **109** are deformed, and the sleeve portion **112** can be inserted into the central channel **108**. After the step portions **122** pass over the top ends of the snap-fit spring clips **109**, the snap-fit spring clips **109** are no longer subjected to the push force and return to their original shape, and the inclined bottom end surfaces **241** of the step portions **122** abut against the inclined top end surfaces **331** of the snap-fit spring clips **109**. The step portions **122** and the shoulder **227** of the second connecter **104** jointly retain the first connecter **102** in position relative to the second connecter **104**.

[0084] It should be noted that the first connecter **102** and second connecter **104** of the fastening clip **100** are connected together when the fastening clip assembly **550** is assembled for the first time. The fastening clip assembly **550** in an assembled state can be obtained by assembling the first component **551** with the first connecter **102** and assembling the second component **552** with the second connecter **104**, respectively. After the first assembly of the fastening clip assembly **550** is completed, the disassembly or assembly of the fastening clip assembly **550** is performed by assembling and disassembling the first connecter **102** and the second connecter **104** of the fastening clip **100**.

[0085] FIGS. **7A-11D** show the specific structures of a fastening clip **700** and a fastening clip assembly **1050** according to another embodiment of the present disclosure.

[0086] FIGS. **7A-7E** show the structure of the fastening clip **700** according to another embodiment of the present disclosure. FIG. **7A** shows a perspective view of the fastening clip **700** at one angle, FIG. **7B** shows an exploded view of the fastening clip **700** at another angle, FIG. **7C** shows a top view of the fastening clip **700**, FIG. **7D** shows a front view of the fastening clip **700**, and FIG. **7E** shows a cross-sectional view of the fastening clip **700** along line E-E. As shown in FIGS. **7A-7E**, similar to the fastening clip **100**, the fastening clip **700** also includes a first connecter **702** and a second connecter **704** that are releasably connected together. The first connecter **702** is configured to connect with a first component **1051** to be securely connected, and a second connecter **704** is configured to connect with a second component **1052** to be securely connected (see FIGS. **10**A and **10**B).

[0087] Similar to the first connecter **102**, the first connecter **702** also includes a neck **705**, a top flange **701** and a bottom flange **703** with a groove **706** formed between the top flange **701** and the bottom flange **703**. A central channel **708** extending along an axis i is defined in the interior of the first connecter **102**. Moreover, at least one snap-fit spring clip **709** and at least one guide strip **707** are connected to an inner wall of the central channel **708**.

[0088] Unlike the first connecter **102**, however, the first connecter **702** and the first component **1051** include a first anti-rotation structure to prevent the relative rotation between them. After the first connecter **702** and the second connecter **704** are connected to each other, it is possible to prevent the rotation of the entire fastening clip relative to the first component **1051** since the relative rotation between the first connecter **702** and the first component **1051** are limited. In this embodiment, the first anti-rotation structure includes a top flange **701** and a square ring-shaped wall **1056**. The top flange **701** is in the shape of a polygon with edges and corners rather than a circular shape, and the square ring-shaped wall **1056** in the shape of a square ring. After the groove **706** of the first connecter **702** receives a flat plate portion **1055** of the first component **1051**, the polygonal top flange **701** can cooperate with the square ring-shaped wall **1056** to prevent the relative rotation between the first connecter **702** and the first component **1051** (see FIG. **10**A). It can be understood by those skilled in the art that the first anti-rotation structure may include other anti-rotation structures, and the top flange **701** and the square ring-shaped wall **1056** may be configured to have any matching shapes as long as the relative rotation between the first connecter **702** and the first component **1051** can be prevented.

[0089] The first connecter **702** further includes a barrel portion **761**. The barrel portion **761** extends axially downwardly and flare outwardly from the bottom of the neck **705**, forming a tapered shape. In this embodiment, the size of the opening at the bottom of the barrel portion **761** is greater than the size of the central channel **708**. In this connection, the second connecter **704** can be inserted more easily into the central channel **708** with the guidance of the barrel portion **761**. Also, the guiding distance during insertion is increased, so that the operator can also connect the first connecter **702** to the second connecter **704** by means of blind insertion in case of a narrow field of view.

[0090] Moreover, the first connecter **702** and the second connecter **704** include a second anti-rotation structure for preventing the relative rotation between them. In this embodiment, the second anti-rotation structure includes a first toothed portion **762** disposed on the first connecter **702** and a second toothed portion **763** disposed on the second connecter **704**. The first toothed portion **762** and the second toothed portion **763** mesh with each other to prevent the relative rotation between the first connecter **702** and the second connecter **704**. In this embodiment, the first toothed portion **762** is disposed at the bottom edge of the barrel portion **761**. The second toothed portion **763** is in the shape of a triangular tooth so that even if the second toothed portion **763** of the second connecter **704** is not fully aligned with the first toothed portion **762** of the first connecter **702**, the triangular tooth can guide them to rotate relative to each other by a small distance so that the two toothed portions mesh with each other.

[0091] Similar to the second connecter **104**, the second connecter **704** includes a main body portion **711**, a sleeve portion **712**, and a flared mouth portion **723**. The sleeve portion **712** is connected to the top of the main body portion **711** and has a step portion **722** at the top end. The step portion **722** is configured to cooperate with the snap-fit spring clips **709** of the first connecter **702**. The flared mouth portion **723** is connected to the bottom of the main body portion **711** and is provided with at least one notch **724** facilitating the deformation of the flared mouth portion **723**. Moreover, the main body portion **711** includes a pair of connection beams **713** and a pair of support beams **714**. The main body portion **711** includes a pair of retaining arms **717** in the interior of the main body portion **711**. Similarly, in some embodiments, the second connecter **704** may alternatively include no flared mouth portion **723**, and the flared mouth portion **723** may alternatively include no notch **724**.

[0092] In this embodiment, however, the sleeve portion **712** does not include the three elastic legs, but is configured to have a complete cylindrical shape. In this embodiment, the sleeve portion **712** is not elastically deformed, and only the snap-fit spring clips **709** will be deformed. This can avoid such a situation that the sleeve portion **712** cannot be elastically deformed inwardly if the length of the pin **1053** of the second component **1052** is relatively long. In this embodiment, the sleeve portion **712** will not affect the insertion force and the retaining force no matter how long the pin **1053** since only the snap-fit spring clips **709** will be deformed. Moreover, the main body portion **711** is not in the shape of a cylinder, but includes a first main body **767** and a second main body **768** connected to each other. The sleeve portion **712** is connected to the top of the first main body **767**, and the flared mouth portion **723** is connected to the bottom of the second main body **768**. The first main body **767** matches the shape of the barrel portion **761** of the first connecter **702**, and extends downwardly and flares outwardly from the bottom of the sleeve portion **712** to form a tapered shape. Moreover, in this embodiment, a pair of support bosses **765** are provided on the outer sides of a pair of connection beams **713** of the second main body **768**. The second toothed portion **763** is arranged on one of the support bosses **765** and is formed protruding upwardly from the top of the support boss **765**. In this way, when the first connecter **702** and the second connecter **704** are connected to each other, the barrel portion **761** is sleeved outside the first main body **767**, and the first toothed portion **762** of the first connecter **702** can mesh with the second toothed portion **763** of the second connecter **704**, so as to prevent the relative rotation between the first connecter **702** and the second connecter **704**. As a specific example, the first toothed portion **762** of the first connecter **702** is configured as a ring of teeth arranged in a circumferential direction, while the second toothed portion **763** of the second connecter **704** is configured as one independent tooth. In this way, even if the first connecter **702** and the second connecter **704** have machining tolerance, the first toothed portion **762** and the second toothed portion **763** can mesh and cooperate with each other more easily. Still referring to FIG. 7E, in this embodiment, the step portion **722** of the sleeve portion **712** has an inclined bottom end surface **741** at its bottom end and a guided surface **742** at its top end. The inclined bottom end surface **741** is configured to urge the deformation of the snap-fit spring clips **709** when the second connecter **104** is pulling out of the first connecter **102** in the top-bottom direction. The guided surface **742** is configured to urge the deformation of the snap-fit spring clips **709** when the second connecter **704** is being inserted into the first connecter **702** in the bottom-top direction. Moreover, in this embodiment, the snap-fit spring clips **709** do not include an inclined top end surface that matches the inclined bottom end surface **741**, but include a flat top end surface **731** which has a smooth edge. The insertion force required for inserting the second connecter **704** into the first connecter **702** and pull-out force (or retaining force) required for pulling the second connecter **704** out of the first connecter **702** can be adjusted by adjusting the inclination angle and the shape of the top ends of the snap-fit spring clips **109**. The applicant has found that, with exerting a certain insertion force and pull-out force, the snap-fit spring clips **709** including the flat top end surface **731** can be repeatedly inserted and pulled out for more times, and the service life of the snap-fit spring clips **709** is longer.

[0093] FIGS. **8**A and **8**B show a more specific structure of the first connecter **702**. FIG. **8**A shows a perspective view of the first connecter **702**, and FIG. **8**B shows a cross-sectional view of the first connecter **702** along line E-E of FIG. **7**C. As shown in FIGS. **8**A and **8**B, the first connecter **702** includes four snap-fit spring clips **709** and four guide strips **707**. Each guide strip **707** has the same shape as the guide strip **107**, and the four guide strips **707** are evenly disposed on the inner wall of the central channel **708**. Each snap-fit spring clip **709** is disposed between two adjacent guide strips **707**. However, the shape of the snap-fit spring clips **709** is not exactly the same as the shape of the snap-fit spring clips **109**.

[0094] Specifically, the bottom end of each snap-fit spring clip **709** is connected to the inner wall of the central channel **708**. Each snap-fit spring clip **709** extends from the bottom end to the top end (which forms a free end) obliquely toward the axis. Each snap-fit spring clip **709** has a guide surface **832** at the bottom end for cooperating with the step portion **722** of the second connecter **704**. Moreover, each snap-fit spring clip **709** has a flat top end surface **731** at the top end. Moreover, in this embodiment, the width of the bottom end of the snap-fit spring clip **709** is greater than the width of the top end. Compared with the snap-fit spring clips **109**, the snap-fit spring clips **709** require an increased deformation force due to the greater width at the bottom ends of the snap-fit spring clips **709**, but will have a prolonged service life after repeated insertion and pull-out.

[0095] Moreover, in this embodiment, the barrel portion **761** of the first connecter **702** and the guide surfaces **832** of the snap-fit spring clips **709** are inclined at substantially the same angle, so as to guide the sleeve portion **712** of the second connecter **704** to be inserted into the central channel **708**.

[0096] The first toothed portion **762** extends axially downwardly from a bottom end edge of the barrel portion **761** and is a ring of triangular teeth arranged continuously in a circumferential direction. All the teeth have the same shape, and triangular grooves **871** are formed between adjacent teeth for receiving the tooth of the second toothed portion **763**. As described above, the triangular teeth facilitate more accurate meshing of the first toothed portion **762** with the second toothed portion **763**.

[0097] FIGS. **9**A and **9**B show a more specific structure of the second connecter **704**. FIG. **9**A shows a perspective view of the second connecter **704**, and FIG. **9**B shows a left side view of the second connecter **704**. As shown in FIGS. **9**A and **9**B, the second connecter **704** includes a pair of support bosses **765***a* and **765***b*. The support bosses **765***a* and **765***b* are disposed oppositely on the left and right sides of the main body portion **711** of the second connecter **704**, and are formed protruding outwardly from an outer surface of the main body portion **711** on the left and right sides. In this embodiment, the support bosses **765***a* and **765***b* are disposed on the right and left sides of the second main body **768** of the main body portion **711**. In this way, when the first connecter **702** is connected to the second connecter **704**, the first main body **767** of the second connecter **704** is supported on an inner side of the barrel portion **761** of the first connecter **702**, and the support bosses **765***a* and **765***b* can support the first toothed portion **762** of the first connecter **702**.

[0098] The second toothed portion **763** is disposed on the support boss **765***a*. The second toothed portion **763** is formed protruding upwardly from an upper surface of the support boss **765**, and is located in the middle of the support boss **765***a*. The second toothed portion **763** is configured to mesh and cooperate with the first toothed portion **762**. In this embodiment, the second toothed portion **763** is also a triangular tooth having a same shape as the grooves formed between adjacent teeth of the first toothed portion **762**, so that the second toothed portion **763** can cooperate with the first toothed portion **762**.

[0099] FIGS. **10**A and **10**B show the structure of a fastening clip assembly **1050** including the fastening clip **700** in an assembled state. FIG. **10**A shows a perspective view of the fastening clip assembly **1050**, and FIG. **10**B shows a cross-sectional view of the fastening clip assembly **1050** taken along line F-F in FIG. **7**C. As shown in FIGS. **10**A and **10**B, the fastening clip assembly **1050** includes a first component **1051**, a second component **1052**, and the fastening clip **700**. The first component **1051** and the first connecter **702** of the fastening clip **700** are connected together. In this embodiment, the first component **1051** includes a substantially square flat plate portion **1055** and a ring-shaped wall **1056**. The flat plate portion **1055** is provided with an opening **1058** on the front side. The ring-shaped wall **1056** is disposed around a peripheral edge of the flat plate portion **1055**. The flat plate portion **1055** is engaged in the groove **706** of the first connecter **702** through the opening **1058**, and is thus received by the groove **706**. In this embodiment, the flat plate portion **1055** has a substantially square shape, and the ring-shaped wall **1056** correspondingly defines a square ring-shaped recess. Moreover, the size of the flat plate portion **1055** is substantially slightly greater than the size of the top flange **701** of the first connecter **702** such that the polygonal top flange **701** can be accommodated within the area surrounded by the ring-shaped wall **1056**, but cannot rotate. As a result, the first connecter **702** cannot rotate relative to the first component **1051**.

[0100] The second component **1052** and the second connecter **704** of the fastening clip **700** are connected together. In this embodiment, the second component **1052** includes a body **1054** and a pin **1053**. The body **1054** and the pin **1053** are fixedly connected together. The pin **1053** can be inserted into the second connecter **704** in the bottom-top direction, and engage with the retaining arms **717**. Moreover, the flared mouth portion **723** abuts against an upper surface of the body **1054**. In this way, the first component **1051** and the second component **1052** can be securely connected to each other. As an example, the pin **1053** is a weld stud having an external thread that mates with the thread engaging cogs of the retaining arms **717**, and the pin **1053** engages with the thread engaging cogs of the retaining arms **717** by means of the external thread, so that the pin **1053** is firmly connected to the second connecter **704**. In this embodiment, the length of the pin **1053** is substantially the same as the overall height of the second connecter **704**, and the middle section of the pin **1053** is configured to engage with the retaining arms **717**. The top of the pin **1053** is substantially retained in the sleeve portion **712**. It should be noted that an inner wall of the sleeve portion **712** has no teeth for engagement with the external thread of the pin **1053**, and the pin **1053** engages with the second connecter **704** only by the retaining arms **717**. In this embodiment, the second connecter **704** may be fitted with a pin **1053** having a length extending from the bottom of the retaining arms **717** to the top of the sleeve portion **712**.

[0101] The barrel portion **761** of the first connecter **702** is sleeved outside the first main body **767** of the second connecter **704**, and the first toothed portion **762** of the first connecter **702** abuts against the support bosses **765** of the second connecter **704**. Therefore, the second connecter **704** cannot move upwardly relative to the first connecter **702**. Moreover, the step portion **722** of the sleeve portion **712** abuts against the top ends of the snap-fit spring clips **709**, so that the second connecter **704** cannot move downwardly relative to the first connecter **702** when the second connecter **704** is not subjected to a large pull-out force. In addition, the first toothed portion **762** of the first connecter **702** meshes with the second toothed portion **763** of the second connecter **704**, so that the second connecter **704** cannot rotate relative to the first connecter **702**. With such a structure, the relatively stable connection between the first connecter **702** and the second connecter **704** can be maintained, such that the fastening clip **700** is formed. Moreover, since the first connecter **702** can be firmly connected to the first component **1051** and the second connecter **704** can be firmly connected to the second component **1052**, the fastening clip **700** can firmly connect the first component **1051** and the second component **1052** together.

[0102] FIGS. **11**A and **11**B show the structure of the fastening clip assembly **1050** including the fastening clip **700** in a disassembled state. FIG. **11**A shows a perspective view of the fastening clip assembly **1150**, and FIG. **11**B shows a cross-sectional view of the fastening clip assembly **1150** taken along line F-F in FIG. **7**C. As shown in FIGS. **11**A and **11**B, when the operator pulls the first component **1051** upwardly or pulls the second component **1052** downwardly, the fastening clip assembly **1050** can be changed from the assembled state to the disassembled state. At this moment, the inclined bottom end surface **741** of the step portion **722** of the second connecter **704** comes into contact with the flat top end surfaces **731** of the snap-fit spring clips **709** and causes the snap-fit spring clips **709** to be deformed inwardly, so that the sleeve portion **712** can be pulled out of the central channel **708**. Moreover, at this moment, the first connecter **702** and the first component **1051** remain connected together, and the second connecter **704** and the second component **1052** remain connected together, but the first connecter **702** is released from the second connecter **704**, so that the first component **1051** is release from the second component **1052**.

[0103] Similar to the fastening clip assembly **550**, by setting the retaining force between the first component **1051** and the first connecter **702**, the retaining force between the first connecter **702** and the second connecter **704**, and the retaining force between the second connecter **704** and the second component **1052** within proper ranges, it is possible to make the disassembly of the first connecter **702** and the second connecter **704** easier than the disassembly of the other components, thereby avoiding the wear between the fastening clip made of plastic and the first and/or second component made of metal.

[0104] When the operator pushes the first component **1051** or the second component **1052**, the fastening clip assembly **1050** can be changed from the disassembled state to the assembled state. At this moment, the sleeve portion **712** of the second connecter **704** is inserted into the barrel portion **761** of the first connecter **702**. After the guided surface **742** of the step portion **722** comes into contact with the guide surface **832** of the snap-fit spring clips **709**, and the guided surface **742** urges the deformation of the snap-fit spring clips **709**, so that the sleeve portion **712** can be inserted into the central channel **708**. After the step portion **722** passes over the top ends of the snap-fit spring clips **709**, the snap-fit spring clips **709** are no longer subjected to the push force and returns to its original shape, and the inclined bottom end surface **741** of the step portion **722** abuts against the flat top end surfaces **731** of the snap-fit spring clips **709**. The step portion **722** and the support bosses **765** of the second connecter **704** jointly retain the first connecter **102** in position relative to the second connecter **104**.

[0105] The present disclosure provides a fastening clip including a first connecter and a second connecter which are releasably connected to each other. When the fastening clip connects two components, the first connecter and the second connecter are firmly connected to the two components, respectively. Moreover, the two components are released from each other by release the first connecter and the second connecter of the fastening clip of the present disclosure, so that it is possible to avoid wear between the components made of a metal material and the fastening clip made of a plastic material. The fastening clip of the present disclosure is particularly suitable for connecting two components with at least one being made of a metal material.

[0106] In the fastening clip of the present disclosure, the first connecter and the second connecter of the fastening clip have smaller sizes by designing the matching shapes of the sleeve and the snap-fit spring clips while the insertion/pull-out force is still within the required range. The second connecter of the fastening clip of the present disclosure includes the retaining arms which provide a firm connection between the second connecter and the pin having an external thread. This further prevents wear between a metal member and a plastic member when the pin is made of a metal material. Moreover, the fastening clip of the present disclosure is connected to the second component by means of a pin such as a weld stud, thereby eliminating the need for forming a hole in the second component, and thus simplifying the assembly process and reducing the assembly difficulty.

[0107] Moreover, in some embodiments of the fastening clip, the first connecter and the second connecter can be better guided to be assembled together by providing the first connecter and the second connecter with the barrel portion and the first main body that match with each other, being particularly suitable for assembly occasions with poor assembly vision.

[0108] In addition, in some embodiments of the fastening clip, the relative rotation between the first connecter and the second connecter can be prevented by providing the first connecter and the second connecter with cooperating anti-rotation structures, such as toothed portions meshing with each other. Moreover, in some embodiments, the fastening clip can also prevent the relative rotation between the first connecter and the first component.

[0109] While the present method and/or system has been described with reference to certain implementations, it will be understood by those skilled in the art that various changes may be made and equivalents may be substituted without departing from the scope of the present method and/or system. In addition, many modifications may be made to adapt a particular situation or material to the teachings of the present disclosure without departing from its scope. For example, block and/or components of disclosed examples may be combined, divided, re-arranged, and/or otherwise modified. Therefore, the present method and/or system are not limited to the particular implementations disclosed. Instead, the present method and/or system will include all implementations falling within the scope of the appended claims, both literally and under the doctrine of equivalents.

Description of main reference signs:

100	fastening clip	700	fastening clip
101	top flange	701	top flange
102	first connecter	702	first connecter
103	bottom flange	703	bottom flange
104	second connecter	704	second connecter
105	neck	705	neck
106	groove	706	groove
107	guide strip	707	guide strip
108	central channel	708	central channel
109	snap-fit spring clip	709	snap-fit spring clip
111	main body portion	711	main body portion
112	Sleeve portion	712	sleeve portion
113	connection beam	713	connection beam
114	support beam	714	support beam
115	pivot portion	717	retaining arm
117	retaining arm	722	step portion
117a	first retaining arm	723	flared mouth portion
117b	second retaining arm	724	notch
118	retaining cavity	731	flat top end surface
119	thread engaging cog	741	inclined bottom end surface
121	elastic leg	742	guided surface
122	step portion	761	barrel portion
123	flared mouth portion	762	first toothed portion
124	notch	763	second toothed portion
227	shoulder	765	support boss
228	sleeve channel	765a	support boss
241	inclined bottom end surface	765b	support boss
242	guided surface	767	first main body
331	inclined top end surface	768	second main body
332	guide surface	832	guide surface
550	fastening clip assembly	871	triangular groove
551	first component	1050	fastening clip assembly
552	second component	1051	first component
554	body	1052	second component
555	flat plate portion	1054	body
556	ring-shaped wall	1055	flat plate portion
553	pin	1056	ring-shaped wall
1053	pin	1150	fastening clip assembly

What is claimed is:

1. A fastening clip, characterized by comprising:
a first connecter having an axis and a central channel extending along the axis, the first connecter being provided with a groove on an outer peripheral wall; and
a second connecter comprising:
a main body portion provided with a retaining cavity and at least one retaining arm, the at least one retaining arm being connected to an inner surface of the main body portion at one end and extending from the inner surface of the main body portion toward the retaining cavity; and
a sleeve portion connected to the top of the main body portion and receivable in the central channel of the first connecter;
wherein the first connecter and the second connecter are configured to be releasably connected to each other.

2. The fastening clip according to claim 1, wherein the first connecter comprises at least one snap-fit spring clip obliquely extending, from a wall that defines the central channel, toward the central channel, and the sleeve portion of the second connecter is releasably connected to the first connecter via the at least one snap-fit spring clip in a snap-fit manner.

3. The fastening clip according to claim 2,
wherein each of the snap-fit spring clips has a connection end and a free end, one end of each of the snap-fit spring clips being connected to the wall that defines the central channel to form the connection end, and the other end of each of the snap-fit spring clips forming the free end; and
wherein the at least one snap-fit spring clip is configured to be elastically deformable to allow the sleeve portion of the second connecter to enter the central channel and restorable to retain the sleeve portion in the central channel.

4. The fastening clip according to claim 3,
wherein the sleeve portion comprises a step portion formed protruding outwardly from an outer wall of the sleeve portion; and
wherein the sleeve portion is configured such that when the at least one snap-fit spring clip is restored, the step portion abuts against the free end of the at least one snap-fit spring clip to retain the sleeve portion in the central channel.

5. The fastening clip according to claim 4,
wherein the step portion has a guided surface disposed at the top of the step portion and extending obliquely outwardly in the top-bottom direction, the guided surface being configured to cooperate with the at least one snap-fit spring clip so as to urge the at least one snap-fit spring clip to elastically deform to allow the sleeve portion of the second connecter to enter the central channel; and/or
wherein the step portion has an inclined bottom end surface disposed at the bottom of the step portion and extending obliquely inwardly in the top-bottom direction, the inclined bottom end surface being configured to cooperate with the snap-fit spring clip so as to abut against the free end of the snap-fit spring clip upon the snap-fit spring clip restores, to retain the sleeve portion in the central channel.

6. The fastening clip according to claim 4, wherein the step portion has a flat top end surface which has a smooth edge.

7. The fastening clip according to claim 1, wherein the first connecter comprises:
a neck defining the central channel;
a top flange formed extending outwardly from the top of the neck; and
a bottom flange formed extending outwardly from the bottom of the neck;
wherein a ring-shaped groove is defined to form on an outer side of the neck between the top flange and the bottom flange.

8. The fastening clip according to claim 7,
wherein the first connecter further comprises a barrel portion formed extending downwardly and flaring outwardly from the bottom of the neck;
wherein the main body portion of the second connecter comprises a first main body formed extending downwardly and flaring outwardly from the bottom of the sleeve portion; and
wherein the barrel portion and the first main body are matched in shape such that when the first connecter is connected to the second connecter, the first main body abuts against an inner side of the barrel portion.

9. The fastening clip according to claim 8,
wherein the first connecter is configured to be mounted together with a first component; and
wherein a first anti-rotation structure is provided on the first connecter and the first component, the first anti-rotation structure comprising the top flange in the shape of a polygon with edges and corners.

10. The fastening clip according to claim 8, wherein a second anti-rotation structure is provided on the first connecter and the second connecter to prevent the relative rotation between the first connecter and the second connecter.

11. The fastening clip according to claim 10, wherein the second anti-rotation structure comprises a first toothed portion disposed at a bottom end of the barrel portion of the first connecter, and a second toothed portion connected to an outer side of the main body portion of the second connecter, wherein the first toothed portion and the second toothed portion mesh and cooperate with each other to prevent the relative rotation between the first connecter and the second connecter.

12. The fastening clip according to claim 11, wherein the first toothed portion and the second toothed portion are triangular teeth.

13. The fastening clip according to claim 12, wherein the first toothed portion comprises a ring of teeth arranged in a circumferential direction, and the second toothed portion comprises a single tooth.

14. The fastening clip according to claim 13, wherein the second connecter further comprises at least one support boss disposed on the outer side of the main body portion and configured to be capable of at least partially supporting the first toothed portion of the first connecter when the first connecter is connected to the second connecter.

15. The fastening clip according to claim 2, wherein the sleeve portion comprises at least one elastic leg configured to be releasably connected to the at least one snap-fit spring clip in a snap-fit manner.

16. A fastening clip assembly, characterized by comprising:

a first component;

a second component comprising a body and a pin which are fixedly connected to each other; and

a fastening clip according to claim **1**;

wherein the first component is received by the groove of the first connecter, and the pin is capable of being cooperatively connected to the at least one retaining arm of the second connecter of the fastening clip to securely connect the first component to the second component.

17. The fastening clip assembly according to claim **16**,

wherein the first component comprises a flat plate portion and a ring-shaped wall arranged around a peripheral edge of the flat plate portion, the flat plate portion being engaged within the groove of the first connecter; and

wherein a first anti-rotation structure is provided on the first connecter and the first component, the first anti-rotation structure comprising the top flange in the shape of a polygon with edges and corners and a ring-shaped wall matching the shape of the top flange, to prevent the relative rotation between the first connecter and the first component.

18. The fastening clip assembly according to claim **16**, wherein one of the first component and the second component is made of a metal material and the other is made of a plastic material.

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